

The Lower Urinary Tract and Male Genital System

George Jabboure Netto and Mahul B. Amin

CHAPTER CONTENTS

The Lower Urinary Tract, 867

Ureters, 868

- Congenital Anomalies, 868
- Tumors and Tumor-like Lesions, 868
- Obstructive Lesions, 868
 - Sclerosing Retroperitoneal Fibrosis*, 868

Urinary Bladder, 868

- Congenital Anomalies, 868
- Inflammation, 869
 - Acute and Chronic Cystitis*, 869
 - Special Forms of Cystitis*, 869
- Metaplastic Lesions, 870
- Neoplasms, 870
 - Urothelial Neoplasms*, 871
 - Other Epithelial Bladder Tumors*, 874
 - Mesenchymal Tumors*, 875
 - Secondary Tumors*, 875
- Obstruction, 875

Urethra, 876

- Inflammation, 876
- Tumors and Tumor-like Lesions, 876

The Male Genital Tract, 876

Penis, 876

- Congenital Anomalies, 876
 - Hypospadias and Epispadias*, 876
 - Phimosis*, 876
- Inflammation, 876
- Tumors, 877
 - Benign Tumors and Tumor-like Conditions*, 877
 - Malignant Tumors*, 877

Testis and Epididymis, 879

- Congenital Anomalies, 879
 - Cryptorchidism*, 879
- Regressive Changes, 880
 - Atrophy and Decreased Fertility*, 880
- Inflammation and Infections, 880
 - Nonspecific Epididymitis and Orchitis*, 880
 - Granulomatous (Autoimmune) Orchitis*, 880
 - Infections*, 880
- Vascular Disorders, 880
 - Torsion*, 880
- Spermatic Cord and Paratesticular Tumors, 881
- Testicular Tumors, 881
 - Germ Cell Tumors*, 881
 - Tumors of Sex Cord–Gonadal Stroma*, 886
 - Gonadoblastoma*, 886
 - Testicular Lymphoma*, 886
- Lesions of Tunica Vaginalis, 886

Prostate, 887

- Inflammation, 887
- Benign Enlargement, 888
 - Benign Prostatic Hyperplasia*, 888
- Neoplasms, 889
 - Adenocarcinoma*, 889
 - Miscellaneous Tumors*, 894

THE LOWER URINARY TRACT

The renal pelves, ureters, urinary bladder, and urethra (except the terminal portion) are lined by a special type of transitional epithelium called *urothelium*. Urothelium is composed of five to six layers of cells with oval nuclei, often with linear nuclear grooves, and a surface layer consisting of large, flattened “umbrella cells” with abundant cytoplasm. This epithelium rests on a basement membrane, beneath which is a lamina propria. The lamina propria in the bladder contains wisps of smooth muscle that form a discontinuous muscularis mucosa. It is important to differentiate the muscularis mucosa from the deeper, larger muscle bundles of the detrusor muscle (muscularis propria), since bladder cancers are staged based on invasion of the muscularis

propria. If urine flow is obstructed and intravesical pressures rise, the bladder musculature undergoes hypertrophy.

The ureters lie throughout their course in the retroperitoneum. Retroperitoneal tumors or fibrosis may entrap and obstruct the ureters. In middle-aged and older females, relaxation of pelvic support leads to prolapse (descent) of the uterus, pulling with it the floor of the bladder. As a result, the bladder protrudes into the vagina, creating a pouch (*cystocele*) that fails to empty readily with micturition. In males, the seminal vesicles and prostate have similar close relationships, being situated just posterior and inferior to the neck of the bladder. Thus, enlargement of the prostate, so common in middle to later life, constitutes an important cause of urinary tract obstruction.

URETERS

Congenital Anomalies

Congenital anomalies of the ureters are found in about 2% to 3% of all autopsies. These include ureteral duplication, vesicoureteral reflux, double and bifid ureters, ureteropelvic junction obstruction, and ureteral diverticula. Although most have little clinical significance, certain anomalies may lead to obstruction of the flow of urine and thus cause clinical disease.

Ureteropelvic junction (UPJ) obstruction is the most common cause of hydronephrosis in infants and children. Cases that present early in life preferentially affect males, are bilateral in 20% of cases, and are often associated with other anomalies. There is agenesis of the contralateral kidney in a minority of cases. In adults, UPJ obstruction is more common in females and is most often unilateral. The condition is most often ascribed to abnormal organization of smooth muscle bundles or excess stromal deposition of collagen between smooth muscle bundles at the UPJ, or in rare cases to extrinsic compression of the UPJ by abnormal renal vessels.

Tumors and Tumor-like Lesions

Primary tumors of the ureter are rare. Benign tumors are generally of mesenchymal origin. *Fibroepithelial polyp*, a tumor-like lesion often occurring in children, is composed of loose, vascularized connective tissue overlaid by urothelium. Primary malignant tumors of the ureter resemble those arising in the renal pelvis, calyces, and bladder. The majority are *papillary urothelial carcinomas* (Fig. 21.1). They occur most frequently during the sixth and seventh decades of life and cause obstruction of the ureteral lumen. They commonly occur concurrently with urothelial carcinomas of the bladder or renal pelvis.

Obstructive Lesions

Intrinsic and extrinsic lesions may obstruct the ureters and may give rise to *hydroureter*, *hydronephrosis*, and *pyelonephritis* (Chapter 20). Unilateral obstruction typically results from proximal intrinsic or extrinsic causes (e.g., stones, neoplasms), whereas bilateral obstruction arises from distal causes, such as nodular hyperplasia of the prostate (Table 21.1).

Sclerosing Retroperitoneal Fibrosis

This rare lesion is characterized by a fibrotic proliferative inflammatory process that encases retroperitoneal structures and causes hydronephrosis. The disorder occurs in middle to late age and is more common



Fig. 21.1 Papillary urothelial cell carcinoma extensively involving the ureter. (Courtesy Dr. Cristina Magi-Galluzzi, The Johns Hopkins Hospital, Baltimore.)

TABLE 21.1 Major Causes of Ureteral Obstruction

Cause of Obstruction	Characteristics/Mechanisms
Intrinsic	
Calculi	Typically of renal origin Usually small (5 mm in diameter or less) Most commonly obstruct at the ureteropelvic junction, where ureters cross iliac vessels, and where the ureter enters the bladder
Strictures	May be congenital or acquired
Tumors	Urothelial carcinomas arising in ureters Rarely, benign tumors or fibroepithelial polyps
Blood clots	Massive hematuria from renal calculi, tumors, or papillary necrosis
Neurogenic	Interruption of the neural pathways to the bladder
Extrinsic	
Pregnancy	Physiologic relaxation of smooth muscle or pressure on ureters at pelvic brim from enlarging fundus
Periureteral inflammation	Salpingitis, diverticulitis, peritonitis, sclerosing retroperitoneal fibrosis
Endometriosis	With pelvic lesions associated with scarring
Tumors	Cancers of the rectum, bladder, prostate, ovaries, uterus, cervix; lymphomas, sarcomas

in males than females. At least a subset of these cases is related to *IgG4-related disease*, a recently described entity associated with elevated levels of serum immunoglobulin G4 (IgG4) and fibroinflammatory lesions rich in IgG4-secreting plasma cells, that also affects other organs (Chapter 6). Other etiologies for retroperitoneal fibrosis include drug exposures (ergot derivatives, β -adrenergic blockers), inflammatory conditions (vasculitis, diverticulitis, Crohn disease), and malignancies (lymphomas, urinary tract carcinomas). Most, however, have no obvious cause and are considered primary or idiopathic (*Ormond disease*).

Microscopic examination typically reveals storiform fibrous tissue containing a prominent infiltrate of lymphocytes, often with germinal centers, plasma cells (frequently IgG4-positive), and eosinophils. Treatment initially involves corticosteroids, although many patients eventually become resistant and require ureteral stents or surgical extrication of the ureters from the surrounding fibrous tissue (*ureterolysis*).

KEY CONCEPTS

Disorders of the Ureters

- Ureteral obstruction is clinically significant because it can lead to hydroureter, hydronephrosis, or pyelonephritis, thus compromising renal function.
- In children, congenital UPJ obstruction is the most common obstructive lesion.
- In adults, ureteral obstruction may be acute (e.g., due to obstructing calculi) or chronic (e.g., due to intrinsic or extrinsic tumors or rarely idiopathic conditions such as sclerosing retroperitoneal fibrosis).

URINARY BLADDER

Congenital Anomalies

Several different types of congenital abnormalities of the bladder are recognized; these are variously associated with increased risk of infection or neoplasia.

- *Vesicoureteral reflux* is the most common and serious congenital anomaly. As discussed in Chapter 20, vesicoureteral reflux predisposes to ascending pyelonephritis and loss of renal function. Abnormal connections between the bladder and the vagina, rectum, or uterus may create *congenital vesicouterine fistulae*.
- *Diverticula* are pouch-like invaginations of the bladder wall that range from less than 1 cm to 10 cm in diameter and may be congenital or acquired. *Congenital diverticula* may be due to a focal failure of development of the normal musculature or to a urinary tract obstruction during fetal development. *Acquired diverticula* are most often associated with prostatic hyperplasia, producing urinary outflow obstruction. The resulting increase in intravesical pressure causes outpouching of the bladder wall and the formation of diverticula. Rarely, carcinoma may arise in bladder diverticula; such tumors are on average more advanced in stage due to the thin or absent muscularis propria layer of diverticula.
- *Exstrophy of the bladder* is a developmental failure in the anterior wall of the abdomen and the bladder. As a result, the bladder communicates directly with the abdominal surface (Fig. 21.2). The exposed bladder mucosa may undergo intestinal glandular metaplasia and is subject to chronic infection that often spreads to the upper urinary tract. Exstrophy is associated with an increased risk of adenocarcinoma in the bladder remnant.
- *Urachal anomalies*. The urachal canal connects the fetal bladder with the allantois and normally is obliterated at birth. Rarely, it remains fully or partially patent. The former creates a fistulous urinary tract connection between the bladder and umbilicus. When only a central region of patent urachus persists, a *urachal cyst* lined by urothelial or metaplastic glandular epithelium is formed. Urachal cysts are at increased risk for neoplastic transformation, mostly presenting as adenocarcinomas (see section on Neoplasms, later in the chapter). These account for only a small minority of bladder cancers overall (0.1% to 0.3%) but constitute 20% to 40% of bladder adenocarcinomas.

Inflammation

Acute and Chronic Cystitis

Bacterial pyelonephritis is frequently preceded by infection of the urinary bladder, with retrograde spread of microorganisms into the kidneys and their collecting systems (discussed in Chapter 20).

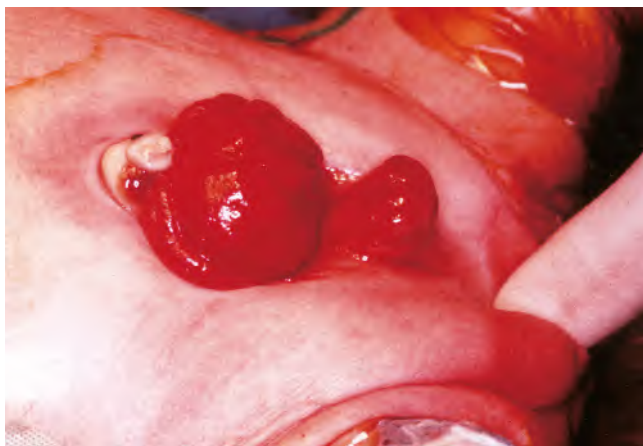


Fig. 21.2 Exstrophy of the bladder in a newborn boy. The tied umbilical cord is seen above the hyperemic mucosa of the everted bladder. Below is an incompletely formed penis with marked epispadias. (Courtesy Dr. John Gearhart, The Johns Hopkins Hospital, Baltimore.)

Predisposing factors for cystitis include bladder calculi, urinary obstruction, diabetes mellitus, instrumentation, and immune deficiency. The most common etiologic agents of cystitis are the coliforms: *Escherichia coli*, followed by *Proteus*, *Klebsiella*, and *Enterobacter*. Females are more likely than males to develop cystitis because of their shorter urethras allow easier access of bacteria from the intestinal tract. *Tuberculous cystitis* is almost always a sequel to renal tuberculosis. *Candida albicans* and, much less often, cryptococcal agents may cause cystitis, particularly in immunosuppressed patients or those receiving long-term antibiotics. Schistosomiasis (*Schistosoma haematobium*) remains an important cause of cystitis in certain African and Middle Eastern countries. Viruses (e.g., adenovirus), *Chlamydia*, and *Mycoplasma* may also cause cystitis. Adenovirus and BK virus infections may result in hemorrhagic cystitis. Gas-forming bacteria (such as *Clostridium perfringens*) result in *emphysematous cystitis* (gas-filled vesicles in the bladder wall).

MORPHOLOGY

Most cases of cystitis produce nonspecific acute or chronic inflammation of the bladder. In **acute cystitis**, there is hyperemia of the mucosa and neutrophilic infiltrate, sometimes associated with exudate. Persistence of the bacterial infection leads to **chronic cystitis** associated with mononuclear inflammatory infiltrates.

Other types of noninfectious cystitis include iatrogenic, follicular, and eosinophilic cystitis. Patients receiving systemic chemotherapy or pelvic irradiation may develop **iatrogenic cystitis**. Cytotoxic agents, such as cyclophosphamide, may cause **hemorrhagic cystitis**. Acute and chronic **radiation cystitis** may occur following the irradiation of the bladder region. **Follicular cystitis** is characterized by the presence of lymphoid follicles within the bladder mucosa and underlying wall.

Clinical Features

All forms of cystitis are characterized by a triad of symptoms: (1) frequency, which in acute cases may necessitate urination every 15 to 20 minutes; (2) lower abdominal pain localized over the bladder region or in the suprapubic region; and (3) dysuria (pain or burning on urination). The local symptoms of cystitis may be merely disturbing, but these infections may be antecedents to pyelonephritis, a more serious disorder (Chapter 20). Cystitis is sometimes a secondary complication of an underlying disorder associated with urinary stasis, such as prostatic hypertrophy, cystocele, calculi, or bladder neoplasms. These primary lesions must be corrected before the cystitis can be relieved.

Special Forms of Cystitis

Several variants of cystitis have distinctive causes or morphologic appearances.

Interstitial Cystitis (Bladder Pain Syndrome). Interstitial cystitis is a disorder of unknown etiology that occurs most frequently in females. The associated pain syndrome is defined by the American Urological Association as “an unpleasant sensation (pain, pressure, discomfort) perceived to be related to the urinary bladder, associated with urinary tract symptoms of more than six weeks duration, in the absence of infection or other identifiable causes.” It is characterized by intermittent, often severe, suprapubic pain; urinary frequency; urgency; hematuria; and dysuria. Typical cystoscopic findings include mucosal fissures and punctate hemorrhages (glomerulations). Microscopically, the pathologic findings are nonspecific; mast cells are often increased in the submucosa, but the pathogenic significance of this finding is uncertain. The main role of biopsy is to rule out carcinoma in situ (CIS), which clinically mimics interstitial cystitis. Treatment is largely empiric. Some cases are associated with chronic mucosal ulcers (*Hunner ulcers*); this is

termed the late (classic, ulcerative) phase. Late in the course, transmural fibrosis may lead to a contracted bladder.

Malakoplakia. Malakoplakia is a distinctive chronic inflammatory reaction that appears to stem from acquired defects in phagocyte function. It arises in the setting of chronic bacterial infection, mostly by *E. coli* or occasionally *Proteus* species, and occurs with increased frequency in the setting of immunosuppression, such as in renal transplant recipients.

MORPHOLOGY

Cystoscopically, malakoplakia takes the form of soft yellow, slightly raised mucosal plaques, 3 to 4 cm in diameter (Fig. 21.3A). Histologically, the lesion is composed of aggregates of large foamy macrophages (**von Hansemann histiocytes**) with occasional multinucleate giant cells and lymphocytes. The **macrophages have abundant granular cytoplasm** due to phagosomes stuffed with particulate and membranous debris of bacterial origin, which are believed to accumulate due to defective phagolysosome function. In addition, laminated mineralized concretions resulting from deposition of calcium in enlarged lysosomes, known as **Michaelis-Gutmann bodies**, are typically present within the macrophages (Fig. 21.3B). Similar lesions have been described in other organs including the colon and other parts of the gastrointestinal tract, brain, lungs, bones, endometrium, kidneys, prostate, and epididymis.

Metaplastic Lesions

Several lesions that stem from true metaplasia or conditions that mimic metaplasia affect the bladder.

- *Cystitis glandularis* and *cystitis cystica*. These are common lesions of the urinary bladder in which nests of urothelium (von Brunn nests) grow downward into the lamina propria. Here, epithelial cells in the center of the nest undergo metaplasia and take on a cuboidal or columnar appearance (*cystitis glandularis*) or retract to produce cystic spaces lined by flattened urothelium (*cystitis cystica*). Because the two processes often coexist, the condition is typically referred to as *cystitis cystica et glandularis*. Both variants can arise in the setting of inflammation and metaplasia. Extensive and multifocal intestinal metaplasia is a precursor to adenocarcinoma.
- *Squamous metaplasia*. As a response to chronic injury, the urothelium is often replaced by nonkeratinizing or keratinizing squamous epithelium, which is a more durable lining. This should be distinguished from glycogenated squamous epithelium that is normally found in females at the trigone region of the bladder. Extensive multifocal keratinizing squamous metaplasia is a precursor to dysplastic lesions and in situ and invasive squamous cell carcinoma. A classic example of this sequence is seen with bladder schistosomiasis, which commonly produces squamous metaplasia and is associated with squamous cell carcinoma in areas in which it is endemic.
- *Nephrogenic adenoma*. Nephrogenic adenoma is an unusual lesion that may not be a form of true metaplasia. There is convincing evidence from renal transplant recipients that at least some of these lesions are caused by implantation and growth of renal tubular cells at sites of bladder mucosa erosion. The overlying urothelium is focally replaced by cuboidal epithelium, which can assume a papillary growth pattern. Although the lesions are typically less than 1 cm in size and asymptomatic, larger lesions have been reported to produce signs and symptoms that raise a suspicion of cancer. In addition, the tubular proliferation can infiltrate the underlying lamina propria and superficial detrusor muscle, microscopically mimicking a malignant process.

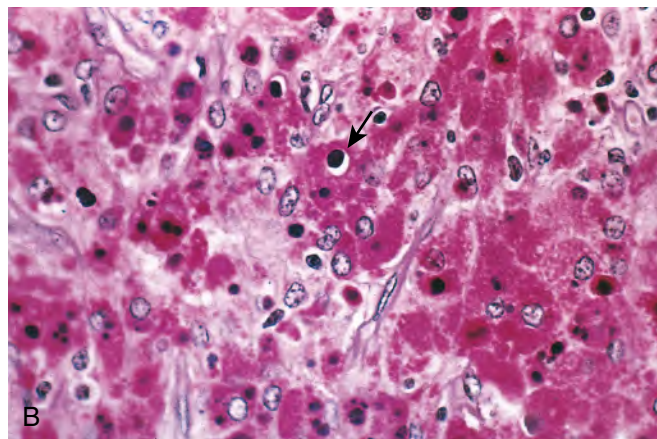
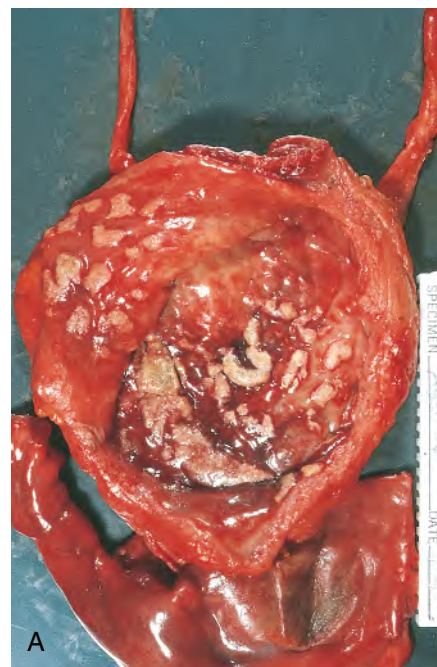


Fig. 21.3 Malakoplakia. (A) Bladder involved by malakoplakia showing the characteristic yellow-orange mucosal lesions. (B) Periodic acid-Schiff stain. Note the large macrophages with granular PAS-positive cytoplasm and several dense, round Michaelis-Gutmann bodies surrounded by artifactual cleared holes in the upper middle field (arrow).

KEY CONCEPTS

Inflammatory Disorders and Metaplasias of the Bladder

- The bladder can be involved by several inflammatory lesions, many of which manifest with frequency and dysuria.
- Acute or chronic bacterial cystitis is extremely common, particularly in females, and results from retrograde spread of colonic bacteria in most cases.
- Other forms of cystitis have iatrogenic causes, such as radiation cystitis and hemorrhagic cystitis due to chemotherapy.
- Some inflammatory or metaplastic bladder lesions are significant in that they may clinically and/or histologically mimic bladder cancer, including malakoplakia, polypoid cystitis, cystitis cystica et glandularis, and nephrogenic adenoma.

Neoplasms

Bladder cancer is the tenth most common cancer type worldwide and is responsible for significant morbidity and mortality. It is the fourth

most common cancer in American males (6% of all new cases). The overwhelming majority (>95%) of bladder tumors are of epithelial origin (Table 21.2), with urothelial neoplasms being by far the most common type, followed by squamous and glandular neoplasms.

Urothelial Neoplasms

Urothelial neoplasms represent about 90% of all bladder tumors and run the gamut from small benign lesions that do not recur to aggressive cancers that are often fatal. Many of these tumors are multifocal at presentation. Though most common in the bladder, all the urothelial lesions described here may be seen at any site where there is urothelium, from the renal pelvis to the distal urethra.

There are two distinct precursor lesions to invasive urothelial carcinoma: noninvasive papillary tumors and flat noninvasive urothelial carcinoma in situ (CIS) (Fig. 21.4). The most common precursor lesions are the noninvasive papillary tumors, which originate from papillary urothelial hyperplasia. These tumors have a range of atypical changes and are graded according to their biologic behavior. The other precursor lesion to invasive carcinoma, flat noninvasive urothelial carcinoma, is referred to as CIS. As discussed in Chapter 7, CIS is a term used to describe epithelial lesions that have the cytologic features of malignancy but are confined to the epithelium, showing no evidence of basement membrane invasion. Such lesions are considered high grade. In about one-half of individuals with invasive bladder cancer, the tumor has already invaded the bladder wall at the time of presentation, and precursor lesions are not detected. It is presumed in such cases that the precursor lesion has been replaced by the high-grade invasive component, which typically appears as a large, ulcerated mass. Although invasion into the lamina propria worsens the prognosis, the major decrease in survival is associated with invasion of the muscularis propria (detrusor muscle). Once muscularis propria invasion occurs, there is a 30% 5-year mortality rate.

Epidemiology. The incidence of bladder cancer is higher in males (male-to-female ratio of 3:1), in higher income nations, and in urban dwellers. About 80% of patients are between 50 and 80 years of age. Bladder cancer, with rare exceptions, is not familial.

Several factors have been implicated in the causation of urothelial carcinoma, including:

- **Cigarette smoking** is clearly the most important influence, increasing the risk three-fold to seven-fold, depending on the duration and type of tobacco use. Between 50% and 80% of all bladder cancers among males are associated with the use of cigarettes. Cigars, pipes, and smokeless tobacco are associated with a smaller risk.
- **Industrial exposure to aryl amines**, particularly 2-naphthylamine and related compounds, as pointed out in our earlier discussion of chemical carcinogenesis (Chapter 7). Cancer appears 15 to 40 years after the first exposure.

TABLE 21.2 Tumors of the Urinary Bladder

Urothelial tumors
Noninvasive urothelial tumors
Invasive urothelial carcinoma
Subtypes: nested, large nested, microcystic, tubular, micropapillary, plasmacytoid, sarcomatoid, giant cell, poorly differentiated, lipid-rich, and glycogen rich (clear cell)
Adenocarcinoma
Squamous cell carcinoma
Mixed carcinoma
Small-cell carcinoma
Mesenchymal tumors

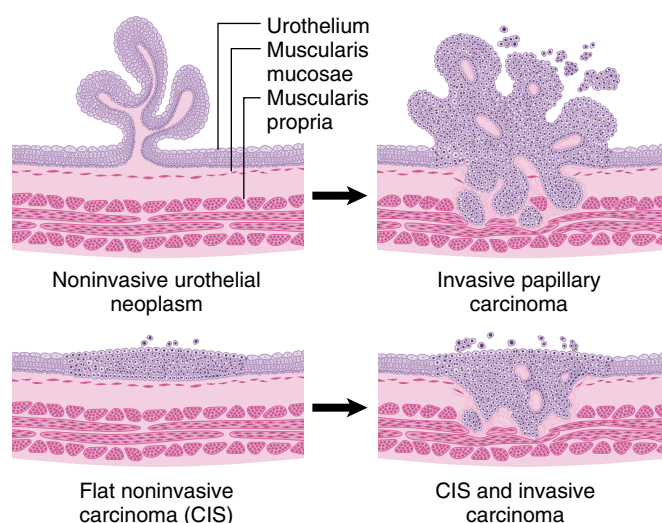


Fig. 21.4 Four morphologic patterns of bladder tumors. *CIS*, Carcinoma in situ.

- *Schistosoma haematobium* infections in endemic areas (Egypt, Sudan) are an established risk. The ova are deposited in the bladder wall and incite a chronic inflammatory response that induces progressive mucosal squamous metaplasia, dysplasia, and, in some instances, neoplasia. Seventy percent of the cancers are squamous, the remainder being urothelial or glandular (the least common type).
- *Long-term use of analgesics* is implicated, as it is in analgesic nephropathy (Chapter 20).
- *Heavy long-term exposure to cyclophosphamide*, an immunosuppressive agent, induces, as noted, hemorrhagic cystitis and increases the risk of bladder cancer.
- *Irradiation*, often administered for other pelvic malignancies (e.g., prostate and uterine cancer), increases the risk of urothelial carcinoma occurring several years following exposure.

Pathogenesis

Analysis of the genomes of precursor lesions and invasive bladder carcinomas has identified two largely distinct molecular pathways of tumor progression (Fig. 21.5). Non-muscle-invasive papillary cancers often have gain-of-function alterations that increase signaling through growth factor receptor pathways, such as amplifications of the *FGFR3* tyrosine kinase receptor gene and activating mutations in the genes encoding RAS and PI 3-kinase. These tumors frequently recur but progress to muscle-invasive bladder cancer in only about 20% of cases. Most muscle-invasive bladder cancers develop by progression from “flat” CIS. Mutations that disrupt the function of p53 and RB are prevalent in all muscle-invasive cancers but occur early in the development of CIS and later in the progression of papillary cancers.

Additional work is ongoing to further characterize the genetic and molecular landscape of bladder cancer. Consistent with epidemiologic data suggesting that environmental carcinogens have an important role, bladder cancers have a high burden of somatic mutations, comparable to other carcinogen-induced cancers, such as lung cancer and melanoma. As in other cancers, recurrently mutated oncogenes and tumor suppressor genes include genes involved in cell cycle regulation, chromatin regulation, DNA repair, and growth factor signaling pathways. Analyses of RNA expression suggest the existence of several relatively distinct molecular subtypes.

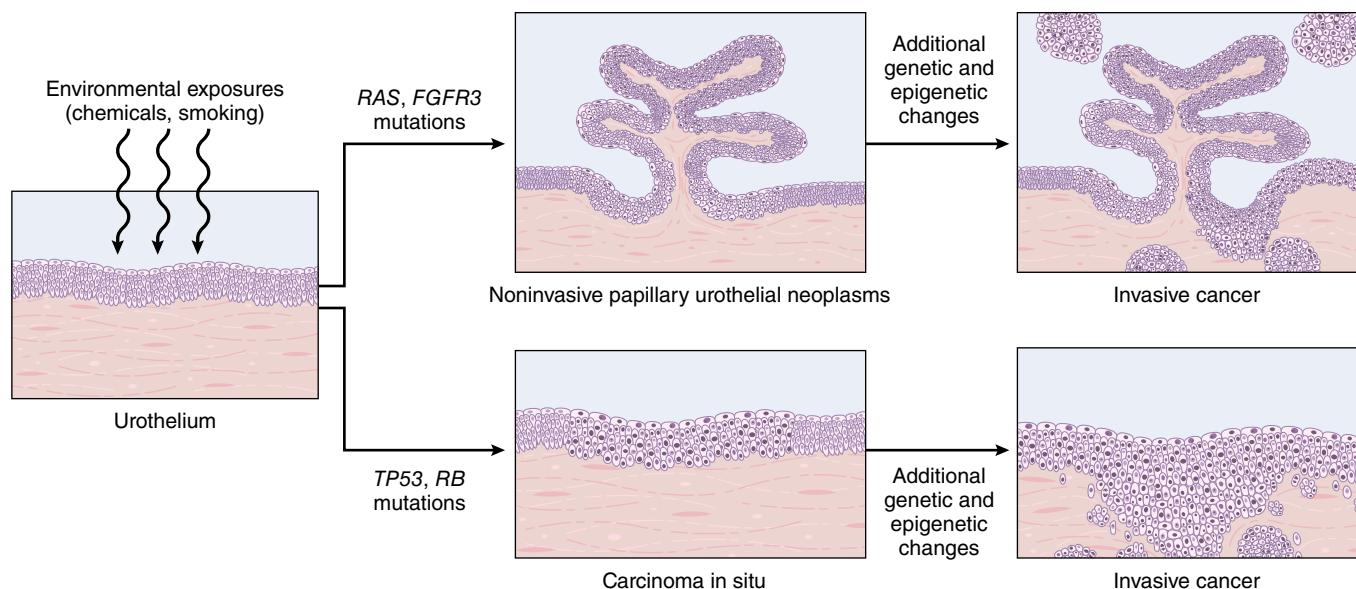


Fig. 21.5 Salient environmental risk factors and major molecular pathways of bladder cancer development from carcinoma in situ and papillary bladder neoplasms. See text for details.

MORPHOLOGY

The appearance of urothelial tumors varies from purely papillary to nodular or flat. Papillary lesions are red, elevated excrescences ranging in size from less than 1 cm in diameter to large masses up to 5 cm in diameter (Fig. 21.6). Multiple discrete tumors are often present.

Table 21.3 lists the classification system of urothelial tumors. **Papillomas** represent 1% or less of bladder tumors and are often seen in younger patients. These tumors typically arise singly as small (0.5 to 2 cm), delicate structures superficially attached to the mucosa by a stalk and are referred to as **exophytic papillomas**. The individual finger-like papillae have a central core of loose fibrovascular tissue covered by epithelium that is histologically identical to normal urothelium (Fig. 21.7A). Recurrences and progression are rare but may occur. In contrast to exophytic papillomas, **inverted papillomas** are completely benign lesions consisting of inter-anastomosing cords of cytologically bland urothelium that extend down into the lamina propria; they simulate an invasive process.

Papillary urothelial neoplasms of low malignant potential (PUNLMP) share many histologic features with papillomas, differing only in having thicker urothelium with greater density of cells (Fig. 21.7B). At cystoscopy, these tumors tend to be larger than papillomas and may be indistinguishable from papillary cancers. Recurrent tumors usually show the same morphology; progression to tumors of higher grade may occur but is rare.

Noninvasive low-grade papillary urothelial carcinomas have an orderly architectural appearance and low-grade cytologic atypia. The cells are evenly spaced (i.e., maintain polarity) and cohesive. There are scattered hyperchromatic nuclei, infrequent mitotic figures predominantly toward the base, and slight variation in nuclear size and shape (Fig. 21.7C). These low-grade cancers may recur and, infrequently, progress in grade and stage. Only rarely do these tumors pose a threat to life.

Noninvasive high-grade papillary urothelial carcinomas contain dyscohesive cells with large hyperchromatic nuclei, irregular nuclear chromatin, and prominent nucleoli. Some of the tumor cells are highly anaplastic (Fig. 21.7D). Mitotic figures, including atypical ones, are frequent. Architecturally, there is disarray and loss of polarity. As compared to low-grade lesions, these tumors have a much higher incidence of progression to muscle-invasive bladder cancer.

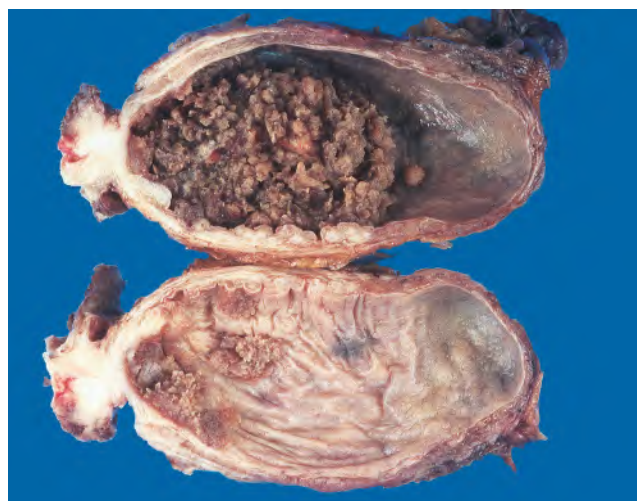


Fig. 21.6 Papillary cancer of the bladder. Cross-section of bladder with the upper section showing a large papillary tumor. The lower section demonstrates multifocal smaller papillary neoplasms. (Courtesy Dr. Fred Gilkey, Sinai Hospital, Baltimore.)

CIS (or flat urothelial carcinoma) is defined by the presence of cytologically malignant cells within a flat urothelium (Fig. 21.8). CIS may range from full thickness cytologic atypia to scattered malignant cells in an otherwise normal urothelium, the latter termed **pagetoid spread**. A common feature shared with high-grade papillary urothelial carcinoma is a lack of cohesiveness, which leads to shedding of malignant cells into the urine. When shedding is extensive, only a few CIS cells may be left clinging to a largely denuded basement membrane. On cystoscopy CIS usually appears as an area of mucosal reddening, granularity, or thickening without an evident intraluminal mass. It is commonly multifocal and may involve most of the bladder surface and extend into the ureters and urethra. **If untreated, 50% to 75% of CIS progresses to invasive cancer.**

Invasive urothelial carcinoma (Fig. 21.9) may be associated with papillary urothelial cancer, usually high grade, or adjacent CIS. The extent of spread

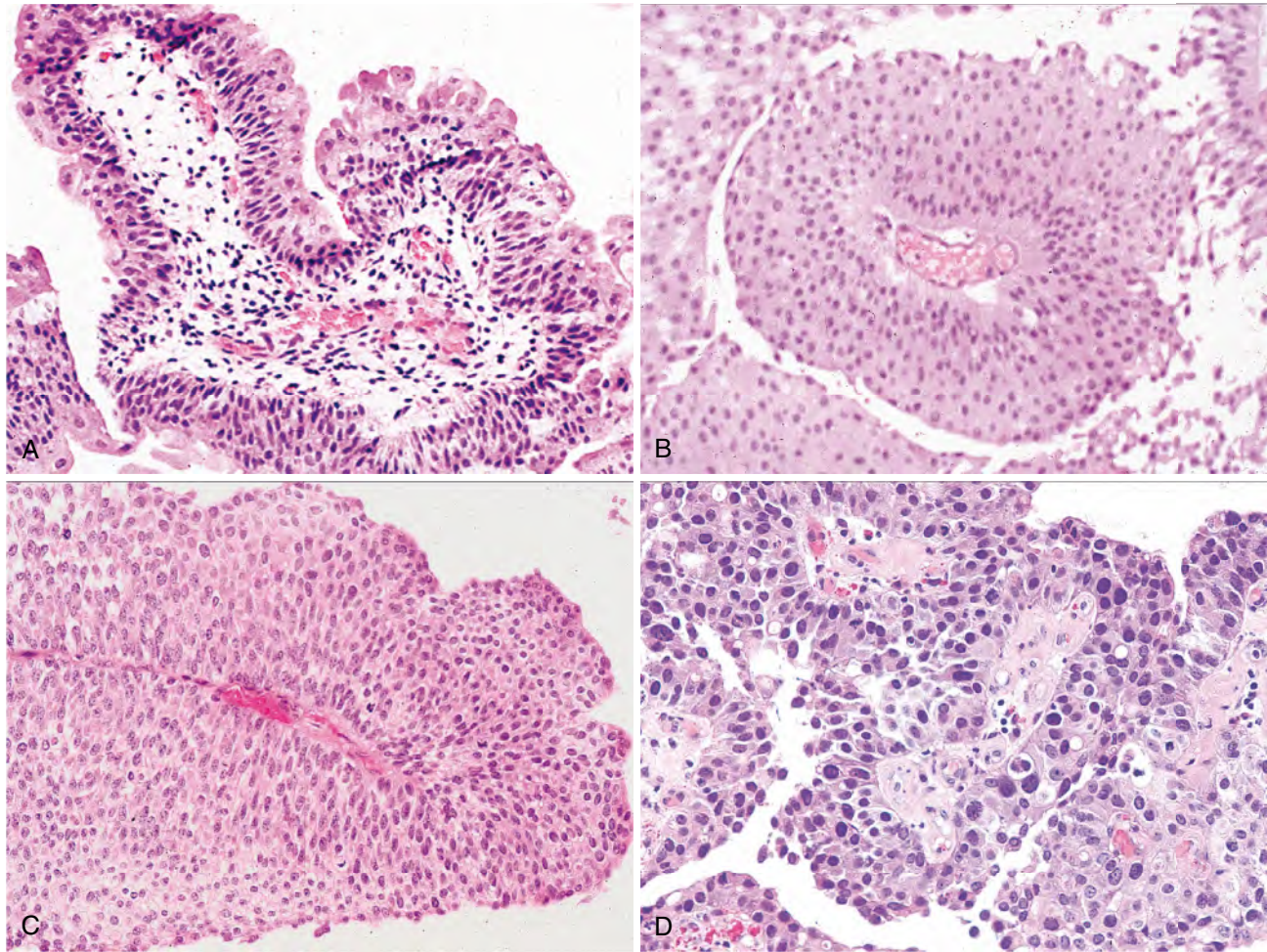


Fig. 21.7 Papillary urothelial neoplasms. (A) Papilloma consisting of small papillary fronds lined by normal-appearing urothelium. (B) Papillary urothelial neoplasms of low malignant potential (PUNLMP) showing thicker urothelium with greater density of cells. (C) Noninvasive low-grade papillary urothelial carcinoma with an overall orderly appearance, with scattered hyperchromatic nuclei and mitotic figures (*upper left*). (D) Noninvasive high-grade papillary urothelial carcinoma with marked cytologic atypia.

(stage), based primarily on depth of invasion in the bladder wall, at the time of initial diagnosis is the most important prognostic factor. Stage also determines treatment modality, with invasion of the muscularis propria layer being an indication for radical cystectomy or radiation therapy with neoadjuvant or adjuvant chemotherapy given the significant potential of “muscle-invasive bladder cancer” for metastasis to regional lymph nodes and systemic spread (e.g., to liver and lung). (Table 21.4).

Unusual subtypes of urothelial cancer include a nested variant with deceptively bland cytology, lymphoepithelioma-like carcinoma, micropapillary carcinoma, and sarcomatoid carcinoma.

Clinical Features

Painless hematuria is the most common symptom of bladder cancer. Frequency, urgency, and dysuria may accompany hematuria. Occasionally, obstruction of the ureteral orifice may lead to pyelonephritis or hydronephrosis.

The subsequent treatment and course of bladder carcinoma depend on tumor grade and stage (Table 21.4), with the key variable being whether the tumor is muscle-invasive or not. Noninvasive papillary

urothelial tumors and those that solely invade the lamina propria (together referred to as non-muscle-invasive) constitute the majority (70% to 80%) of urothelial neoplasms. Between 20% and 30% of bladder cancers are muscle-invasive with only a minority of these originating in patients with a prior history of non-muscle-invasive disease.

Initial treatment of non-muscle-invasive tumors is guided by pathologic findings. For small, localized low-grade papillary tumors, diagnostic transurethral resection is the only procedure needed. CIS and papillary tumors that are large, high grade, multifocal, have a history of recurrence, or are associated with lamina propria invasion are treated with intravesical instillation of an attenuated strain of *Mycobacterium bovis* called bacillus Calmette-Guérin (BCG), which elicits a local inflammatory reaction that destroys the tumor.

As mentioned above, non-muscle-invasive bladder cancers have a high tendency to recur (up to 70%). Most recurrences appear at sites other than that of the original lesion, yet are clonally related, apparently arising from shedding and implantation of cells from the original tumor at a distant site. Other recurrences are clonally distinct and represent a second primary tumor. Papillomas, PUNLMP, and low-grade papillary urothelial cancer have a 98% 10-year survival rate regardless of the number of recurrences, and only a few tumors

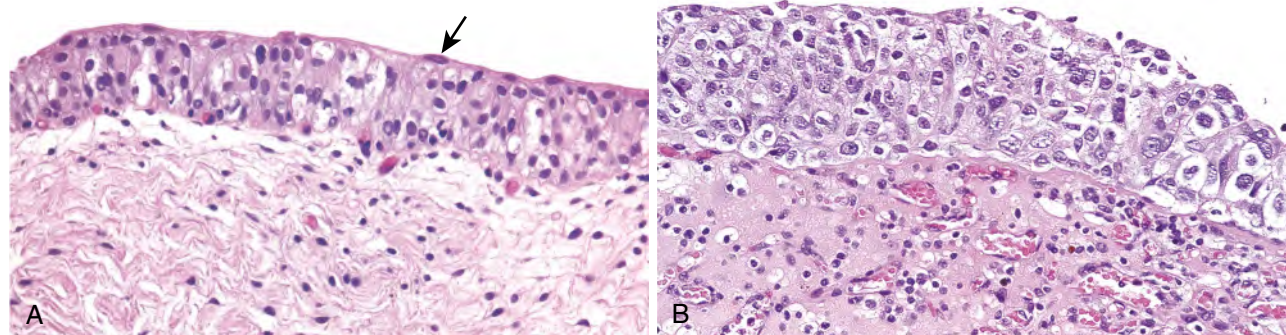


Fig. 21.8 Bladder carcinoma in situ. (A) Normal urothelium with uniform nuclei and well-developed umbrella cell layer (arrow). (B) Carcinoma in situ showing a disorganized urothelium containing numerous cells having markedly enlarged and pleomorphic nuclei.



Fig. 21.9 Opened bladder showing a high-grade invasive urothelial cell carcinoma at an advanced stage. The aggressive multinodular neoplasm has fungated into the bladder lumen and spread over a wide area. The yellow regions represent areas of ulceration and necrosis.

(<10%) progress to higher-grade lesions. In contrast, high-grade papillary urothelial carcinomas invade and lead to death in about 25% of cases. Patients with CIS only, as opposed to CIS associated with infiltrating urothelial carcinoma, are less likely to progress to muscle-invasive cancer (28% versus 59%) or die of disease (7% versus 45%).

Because of the risk of recurrence and progression, patients with non-muscle-invasive urothelial neoplasms need lifelong surveillance

TABLE 21.3 WHO Classification (2022) of Noninvasive Urothelial Tumors

Flat Lesions
Urothelial dysplasia
Urothelial carcinoma in situ (CIS)
Exophytic Papillary Lesions
Papilloma
Papillary urothelial neoplasms of low malignant potential (PUNLMP)
Papillary urothelial carcinoma, low grade
Papillary urothelial carcinoma, high grade

WHO, World Health Organization.

From Netto GJ, Amin MB, Berney DM, et al. The 2022 World Health Organization Classification of Tumors of the Urinary System and Male Genital Organs—Part B: Prostate and Urinary Tract Tumors. *Eur Urol.* 2022;82(5):469-482.

and follow-up. Several noninvasive assays to screen for recurrence have been developed, but none have been proven to have sufficient sensitivity and specificity to replace cystoscopy.

Muscle-invasive bladder carcinoma is treated with radical cystectomy or cystoprostatectomy, or radiation therapy with neoadjuvant and/or adjuvant chemotherapy. In addition to muscle-invasive cancers, radical cystectomy is also indicated in cases of (1) CIS or high-grade papillary cancer refractory to BCG and other intravesical therapies; (2) CIS extending into the prostatic urethra and ducts, sites where instilled BCG does not come into contact with the neoplastic cells; and (3) occasional cases of non-muscle-invasive papillary urothelial high-grade carcinoma, such as multifocal lesions that are too large and extensive to completely eradicate by transurethral resection.

Once bladder carcinoma metastasizes, treatment options are limited. Most metastatic tumors respond poorly to chemotherapy, which produces 5-year survival rates of only 15%. As with other cancers linked to environmental carcinogens with high mutational loads, a subset (roughly 30%) of metastatic bladder carcinomas respond to immune checkpoint inhibitors, sometimes dramatically, providing some hope for this group of patients. Novel targeted therapeutic options for locally advanced and metastatic urothelial carcinomas include a group of “antibody-drug conjugates” directed toward biomarkers expressed in such tumors (e.g., Nectin-4 and FGFR3).

Other Epithelial Bladder Tumors

Squamous cell carcinomas resembling squamous cell cancers occurring at other sites make up 3% to 7% of bladder cancers in the United States but occur much more frequently in countries where urinary schistosomiasis is endemic. Pure squamous cell carcinomas arise

TABLE 21.4 Pathologic Staging of Urinary Bladder Carcinoma (Primary Tumor): pTNM, AJCC 8th Edition

Tumor	
pTX	Primary tumor cannot be assessed
pT0	No evidence of primary tumor
Ta	Noninvasive papillary carcinoma
Tis	Urothelial carcinoma in situ: “flat CIS”
T1	Tumor invades lamina propria (subepithelial connective tissue)
T2	Tumor invades muscularis propria
T2a	Tumor invades superficial muscularis propria (inner half)
T2b	Tumor invades deep muscularis propria (outer half)
T3	Tumor invades perivesical soft tissue
T3a	Tumor invades perivesical soft tissue microscopically
T3b	Tumor invades perivesical soft tissue macroscopically (extravesical mass)
T4	Extravesical tumor directly invades any of the following: prostatic stroma, seminal vesicles, uterus, vagina, pelvic wall, abdominal wall
T4a	Extravesical tumor invades directly into prostatic stroma, seminal vesicles, uterus, or vagina
T4b	Extravesical tumor invades pelvic wall or abdominal wall
Regional Lymph Nodes (pN)	
NX	Lymph nodes cannot be assessed
N0	No lymph node metastasis
N1	Single regional lymph node metastasis in the true pelvis (perivesical, obturator, internal and external iliac, or sacral lymph node)
N2	Multiple regional lymph node metastasis in the true pelvis (perivesical, obturator, internal and external iliac, or sacral lymph node metastasis)
N3	Lymph node metastasis to the common iliac lymph nodes
Metastasis	
M1	Distant metastasis
M1a	Distant metastasis limited to lymph nodes beyond the common iliac arteries
M1b	Non-lymph node distant metastases

AJCC, American Joint Committee on Cancer.

from atypical keratinizing mucosa (squamous dysplasia and CIS) and are nearly always associated with chronic bladder irritation and infection. *Mixed urothelial carcinoma with areas of squamous carcinoma* is more frequent than pure squamous cell carcinoma. Most are invasive, fungating tumors or are infiltrative and ulcerative. The level of cellular differentiation varies widely, from well-differentiated lesions producing abundant keratin to anaplastic tumors with only focal evidence of squamous differentiation. *Adenocarcinoma* of the bladder is rare and histologically identical to adenocarcinomas seen in the gastrointestinal tract.

Small-cell carcinoma, indistinguishable from small-cell carcinomas of the lung, occasionally arises in the bladder, often adjacent to urothelial, squamous, or adenocarcinoma (see Table 21.2). Arising from neuroendocrine cells, it is associated with advanced age and smoking. As with small-cell carcinomas in other anatomic sites, these very aggressive tumors commonly harbor loss-of-function mutations in the *TP53* and *RB* tumor suppressor genes.

Mesenchymal Tumors

Benign Tumors. A great variety of benign mesenchymal tumors may arise in the bladder, having the histologic features of their counterparts elsewhere. Even collectively, they are rare. The most common is *leiomyoma*. They tend to grow as isolated, intramural (submucosal), encapsulated, oval-to-spherical masses up to several centimeters in diameter.

Sarcomas. True sarcomas are distinctly uncommon in the bladder. Inflammatory myofibroblastic tumors and various carcinomas that assume sarcomatoid growth patterns are more common than true sarcomas. As a group, sarcomas tend to produce large masses (up to 15 cm in diameter) that protrude into the vesicle lumen. Their soft, fleshy, gray-white gross appearance suggests their mesenchymal nature. The most common bladder sarcoma in infancy or childhood is *embryonal rhabdomyosarcoma*. In some of these cases they manifest as a polypoid grape-like mass (*sarcoma botryoides*). The most common bladder sarcoma in adults is *leiomyosarcoma* (Chapter 26).

Secondary Tumors

Secondary malignant involvement of the bladder is most often due to direct extension of cancers arising in adjacent organs, mainly the cervix, uterus, prostate, and rectum. Lymphoma may involve the bladder as a component of systemic disease. Metastatic spread of solid tumors to the bladder may occur but is very rare.

KEY CONCEPTS

Bladder Neoplasms

- Bladder cancer is most common in older males, and cigarette smoking constitutes one of the most important risk factors.
- Painless hematuria is a common presenting symptom of bladder cancer and requires clinical investigation by cystoscopy and/or urine cytology analysis to rule out urothelial neoplasia.
- Non-muscle-invasive bladder cancer is associated with gain-of-function mutations in growth factor signaling pathways and frequently recur but rarely progress in stage or grade.
- Muscle-invasive bladder cancers are associated with inactivation of *TP53* and *RB* tumor suppressor genes and often develop from “flat” carcinoma in situ, with or without a high-grade papillary component.
- Other epithelial bladder tumor types may occur, either alone or mixed with urothelial carcinoma, including squamous cell carcinoma, adenocarcinoma, and small-cell carcinoma.

Obstruction

Obstruction of the bladder outlet is of major clinical importance because of its eventual effect on the kidney. In males, the most common cause is enlargement of the prostate gland due to benign prostatic hyperplasia (BPH). Bladder obstruction is less common in females and is most often caused by cystocele of the bladder. Infrequent causes are (1) congenital urethral strictures; (2) inflammatory urethral strictures; (3) inflammatory fibrosis and contraction of the bladder; (4) bladder tumors, either benign or malignant; (5) invasion of the bladder neck by tumors arising in contiguous organs; (6) mechanical obstructions caused by foreign bodies and calculi; and (7) injury of nerves controlling bladder contraction (neurogenic bladder).

MORPHOLOGY

In the early stages there is only thickening of the bladder wall due to smooth muscle hypertrophy. With progressive hypertrophy the individual muscle bundles greatly enlarge and produce trabeculation of the bladder wall (Fig. 21.10). In the course of time, crypts form and may be converted into diverticula.

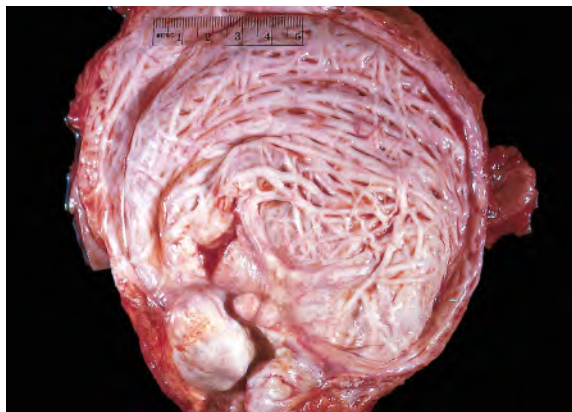


Fig. 21.10 Hypertrophy and trabeculation of bladder wall secondary to benign prostatic hyperplasia.

In some cases of acute obstruction, or in terminal disease when the patient's normal reflex mechanisms are depressed, the bladder may become extremely dilated. The enlarged bladder may reach the brim of the pelvis or even the level of the umbilicus. In such cases, the bladder wall is markedly thinned and lacks trabeculations.

URETHRA

Inflammation

Urethritis is classically divided into gonococcal and nongonococcal causes. *Gonococcal urethritis* is one of the earliest manifestations of this venereal infection. *Nongonococcal urethritis* is common and can be caused by several different organisms. Various strains of *Chlamydia* (e.g., *Chlamydia trachomatis*) are the cause of 25% to 60% of nongonococcal urethritis in males and about 20% in females. *Mycoplasma (Ureaplasma urealyticum)* also accounts for symptoms of urethritis in many cases. Urethritis is often accompanied by cystitis in females and by prostatitis in males. In many instances of suspected bacterial urethritis, no organism can be isolated. Some urethritis is truly noninfectious in origin. An example of such an inflammatory urethritis is a disorder known as *reactive arthritis* (formerly known as Reiter syndrome), which is associated with the clinical triad of arthritis, conjunctivitis, and urethritis (Chapter 26).

The morphologic changes are entirely typical of inflammation in other sites within the urinary tract. The urethral involvement is not itself a serious clinical problem but may cause considerable local pain, itching, and frequency and may warn of more serious disease at higher levels of the urogenital tract.

Tumors and Tumor-like Lesions

Urethral caruncle is an inflammatory lesion that presents as a small, red, painful mass about the external urethral meatus, typically in older females. It consists of inflamed granulation tissue covered by an intact but friable mucosa, which may ulcerate and bleed with the slightest trauma. Surgical excision affords prompt relief and cure.

Benign epithelial tumors of the urethra include squamous and urothelial papillomas, inverted urothelial papillomas, and condylomas. *Primary carcinoma of the urethra* is an uncommon lesion (Fig. 21.11). Cancers arising within the proximal urethra tend to show urothelial differentiation and are analogous to those occurring within the bladder, whereas lesions originating within the distal urethra are more often squamous cell carcinomas and related to human papillomavirus (HPV)

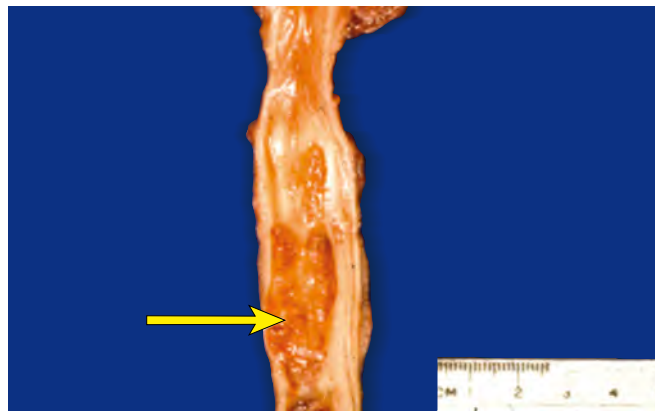


Fig. 21.11 Carcinoma of urethra with typical fungating growth pattern (arrow).

infection. Adenocarcinomas are infrequent in the urethra and generally occur in females. Neoplasms of the prostatic urethra are addressed later in the section on the prostate.

THE MALE GENITAL TRACT

PENIS

Congenital Anomalies

The penis is involved by many congenital anomalies; only those that are clinically significant are discussed here.

Hypospadias and Epispadias

Malformation of the urethral groove and canal may create an abnormal opening either on the ventral surface of the penis (*hypospadias*) or on the dorsal surface (*epispadias*). Either of these anomalies can be associated with failure of normal descent of the testes and with malformations of the urinary tract. Hypospadias, the more common of the two, occurs in approximately 1 in 300 live male births. Even when isolated, these urethral defects may have clinical significance because the abnormal opening is often constricted, resulting in urinary tract obstruction and an increased risk of ascending infections. When the orifices are situated near the base of the penis, normal ejaculation and insemination are hampered and may be a cause of sterility.

Phimosis

When the orifice of the prepuce is too small to permit its normal retraction, the condition is designated *phimosis*. An abnormally small orifice may result from anomalous development but is more frequently the result of repeated bouts of infection that cause scarring of the preputial ring. Phimosis is important because it interferes with cleanliness and permits the accumulation of secretions and detritus under the prepuce, favoring the development of secondary infections and increasing the risk for penile carcinoma.

Inflammation

Inflammations of the penis almost invariably involve the glans and prepuce and include a wide variety of specific and nonspecific infections. The specific infections (i.e., syphilis, gonorrhea, chancroid, granuloma inguinale, lymphopathia venerea, genital herpes) are sexually transmitted and are discussed in Chapter 8. Only the nonspecific infections causing so-called balanoposthitis are described here.

Balanoposthitis refers to infection of the glans and prepuce caused by a wide variety of organisms. Among the more common agents are *C. albicans*, anaerobic bacteria, *Gardnerella*, and pyogenic bacteria. Most cases occur because of poor local hygiene in uncircumcised males, in whom the accumulation of desquamated epithelial cells, sweat, and debris, termed *smegma*, acts as a local irritant. Persistence of such infections leads to inflammatory scarring and, as mentioned earlier, is a common cause of phimosis.

Tumors

Tumors of the penis are uncommon. The most frequent neoplasms are squamous cell carcinoma and benign genital warts (condyloma acuminatum).

Benign Tumors and Tumor-like Conditions

Condyloma Acuminatum. Condyloma acuminatum is a benign sexually transmitted wart caused by human papillomavirus (HPV). It is related to the common wart and may occur on any moist mucocutaneous surface of the external genitals in either sex. “Low risk” HPV serotypes (HPV 6 and, less frequently, HPV 11) are the most frequent cause of condylomata acuminata.

MORPHOLOGY

Condylomata acuminata may arise in the external genitalia or perineal areas. Penile lesions usually occur in the coronal sulcus and inner surface of the prepuce. They consist of **single or multiple sessile or pedunculated, red papillary excrescences** that are several millimeters in diameter (Fig. 21.12). Histologically, a branching, villous, papillary connective tissue stroma is covered by epithelium that may have considerable superficial hyperkeratosis and thickening of the underlying epidermis (**acanthosis**) (Fig. 21.12A). The normal orderly maturation of the epithelial cells is preserved. The lining cells frequently display perinuclear cytoplasmic vacuolization (koilocytosis), characteristic of HPV infection (Fig. 21.12B).

Peyronie Disease. This disorder, which appears to be reactive rather than neoplastic, is characterized by hard penile plaques that result from the deposition of collagen in the connective tissue between the corpora cavernosa and the tunica albuginea. The fibrosis is believed to be the product of microvascular trauma and subsequent

organizing sclerosing chronic inflammation. Clinically, the lesion results in penile curvature toward the side of the lesion and pain during intercourse. Treatments include surgery and injection of collagenase to lyse the fibrous plaques.

Malignant Tumors

Squamous Carcinoma in Situ/Penile Intraepithelial Neoplasia. As at other sites, such as uterine cervix, in situ squamous lesions span a range of morphologies from CIS to less severe derangements. These lesions are encompassed by the umbrella term *penile intraepithelial neoplasia* (PeIN). All are squamous lesions confined to the epidermis by an intact basement membrane. PeIN may be HPV associated or HPV independent. The latter is seen with *balanitis xerotica obliterans*, a chronic inflammatory condition consisting of white patches that occur on the foreskin of older patients, and retains a degree of squamous maturation. HPV-associated PeIN is composed of more overtly malignant cells and manifests clinically as two distinct lesions: Bowen disease and bowenoid papulosis. Both are associated with high-risk HPV, most commonly HPV 16.

- **Bowen disease** most commonly affects the penile shaft and scrotum of older males. At these sites, it appears as a solitary, thickened, gray-white, opaque plaque. In less common cases when it affects the glans, the lesion acquires a velvety red appearance. Histologically, the lesion consists of dysplastic squamous cells containing large hyperchromatic irregular nuclei and lacking orderly maturation (Fig. 21.13). Mitoses, some atypical, are numerous. In 10% of patients, Bowen disease gives rise to infiltrating squamous cell carcinoma.
- **Bowenoid papulosis** occurs in sexually active adults. It is distinguished from Bowen disease by the younger age of affected patients and its presentation as multiple (rather than solitary) reddish brown papular lesions. Although etiologically related to HPV 16 and histologically indistinguishable from Bowen disease, bowenoid papulosis virtually never develops into invasive carcinoma and usually regresses spontaneously.

Invasive Squamous Cell Carcinoma. Squamous cell carcinoma of the penis is associated with poor genital hygiene and high-risk HPV infection. Penile carcinoma affects middle-aged and older patients (40 to 70 years of age). In the United States, it accounts for less than 1% of cancers in males. In contrast, in some parts of Asia, Africa, and South America, penile cancer accounts for 10% to 20% of male

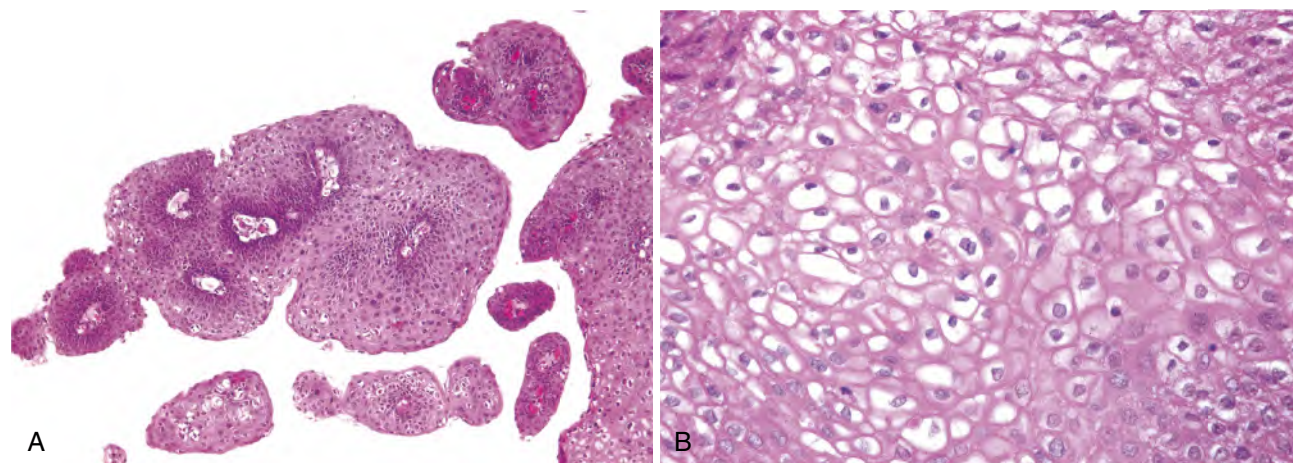


Fig. 21.12 Condyloma acuminatum of the penis. (A) Low magnification reveals the papillary architecture and thickening of the epidermis. (B) On high magnification the epithelium shows perinuclear vacuolization (koilocytosis) characteristic of human papillomavirus infection.

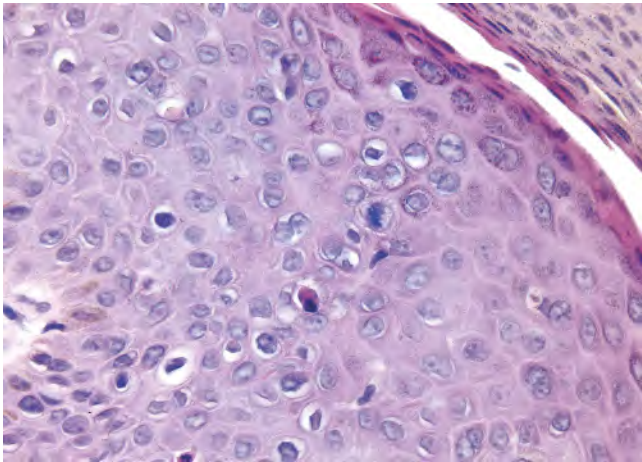


Fig. 21.13 Bowen disease (carcinoma in situ) of the penis. Note the hyperchromatic, dysplastic dyskeratotic epithelial cells and scattered mitoses (mid-right) well above the basal layer.

malignancies. Low-income status and poor hygiene habits are salient risk factors. Circumcision confers protection, and hence penile cancer is extremely rare among Jewish and Muslim people and is correspondingly more common in populations in which circumcision is not practiced routinely. It is postulated that circumcision reduces exposure to carcinogens that may be concentrated in smegma and decreases the likelihood of infections with potentially oncogenic types of HPV. Only a minority of penile cancer is HPV-associated; other factors such as poor hygiene and chronic inflammation presumably contribute to those that are HPV independent. The availability of vaccines to both low-risk and high-risk subtypes of HPV may help reduce the incidence of penile cancer and condyloma acuminatum. Other risk factors include cigarette smoking and chronic inflammatory conditions such as *lichen sclerosis et atrophicus* (*balanitis xerotica obliterans*).

Pathogenesis

The contribution of high-risk HPV to penile carcinoma is identical to its role in cervical cancer (Chapters 7 and 22). These subtypes of HPV encode E6 and E7 proteins that inactivate the p53 and RB tumor suppressor proteins, leading to genomic instability and increased proliferation, respectively. The pathogenesis of HPV-independent penile carcinoma is not fully understood but is associated with chronic inflammatory conditions as well as immunosuppression, such as in patients with HIV or following organ transplant.

MORPHOLOGY

Squamous cell carcinoma of the penis usually originates in the glans or inner surface of the prepuce near the coronal sulcus. Macroscopically the tumors may be irregular, fungating cauliflower-like masses; flat, indurated lesions; or large verruciform/papillary tumors (Fig. 21.14). The World Health Organization (WHO) 2022 classification of penile squamous cell carcinoma recognizes HPV-associated and HPV-independent categories, each with several different subtypes (Table 21.5). Conventional (usual) squamous cell carcinoma is the most common HPV-independent subtype, encompassing almost half of all penile cancers. Pathologic prognostic factors in penile carcinoma include stage, grade, histologic subtype, vascular invasion, and perineural invasion. Many of the histologic subtypes are associated with distinct grades (e.g., verrucous and papillary carcinomas are well differentiated/grade 1 tumors, while sarcomatoid and basaloid carcinomas are poorly differentiated/grade 3 tumors. Among the

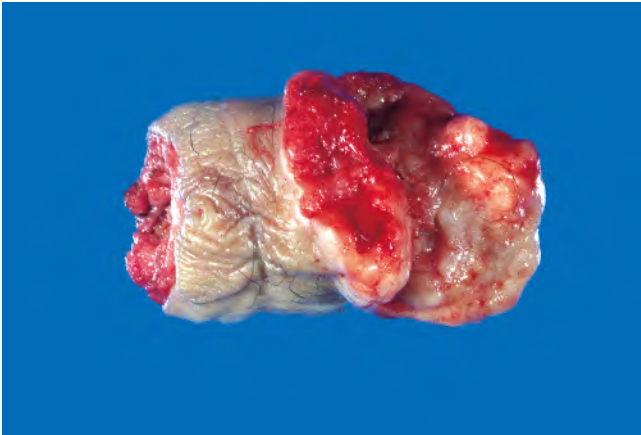


Fig. 21.14 Carcinoma of the penis. The glans penis is deformed by a firm, ulcerated, infiltrative mass.

special subtypes of penile carcinoma, several merit brief description. **Verrucous carcinoma** is an exophytic, warty well-differentiated, HPV-independent subtype that invades locally along a broad pushing border, but rarely metastasizes. In contrast, **basaloid squamous carcinoma** is an HPV-associated subtype composed of relatively small hyperchromatic cells that has a destructive pattern of invasion and usually pursues an aggressive course.

Clinical Features

Invasive squamous cell carcinoma of the penis is a slowly growing, locally invasive lesion that often has been present for a year or more before it is brought to medical attention. The lesions are not painful until they undergo secondary ulceration and infection. Metastases to inguinal lymph nodes may occur early in the disease course, but widespread dissemination is rare until the lesion is far advanced. Clinical assessment of regional lymph node involvement is notoriously inaccurate; only 50% of enlarged inguinal nodes detected in males with penile squamous cell carcinoma contain cancer, with the remainder showing only reactive lymphoid hyperplasia. As mentioned above, prognosis is strongly correlated with tumor stage at diagnosis. Patients with penile cancer that has metastasized to multiple, bilateral lymph or pelvic lymph nodes have a guarded prognosis, with reported 5-year disease-specific survival rates ranging from 7% to 60%.

TABLE 21.5 World Health Organization 2022 Pathologic Classification of Squamous Cell Carcinoma of the Penis

HPV-Independent Squamous Cell Carcinoma
Squamous cell carcinoma, usual type
Verrucous carcinoma (including carcinoma cuniculatum)
Papillary squamous cell carcinoma
Sarcomatoid (spindle) squamous cell carcinoma
HPV-Associated Squamous Cell Carcinoma
Basaloid squamous cell carcinoma
Warty carcinoma
Clear cell squamous carcinoma
Lymphoepithelial carcinoma
Others
Adenosquamous carcinoma
Mucoepidermoid carcinoma
Extramammary Paget disease

HPV, Human papillomavirus.
From Moch H, Amin MB, Berney DM, et al. The 2022 World Health Organization Classification of Tumours of the Urinary System and Male Genital Organs-Part A: Renal, Penile, and Testicular Tumours. *Eur Urol.* 2022;82(5):458-468.

KEY CONCEPTS

Carcinoma of the Penis

- Most common in lower income parts of the world, such as Africa, Asia, and South America, due in part to a lower incidence of circumcision.
- Two pathogenic pathways, one related to HPV and the other unrelated to HPV.
- Squamous cell carcinoma occurs on the glans or shaft of the penis as an ulcerated infiltrative lesion that may spread to inguinal nodes and infrequently to distant sites.
- Other important lesions of the penis include congenital abnormalities involving the position of the urethra (epispadias, hypospadias) and inflammatory disorders (balanitis, phimosis).

TESTIS AND EPIDIDYMIS

Distinct pathologic conditions affect the testis and epididymis. In the epididymis, the most frequent conditions are inflammatory diseases, whereas tumors dominate in the testis.

Congenital Anomalies

Except for undescended testes (cryptorchidism), congenital anomalies are extremely rare and include absence of one or both testes and fusion of the testes (*synorchism*).

Cryptorchidism

Cryptorchidism is a complete or partial failure of the intraabdominal testes (one or both) to descend into the scrotal sac and is associated with testicular dysfunction and an increased risk of testicular cancer. It is found in approximately 1% of 1-year-old boys. It usually occurs as an isolated anomaly but may be accompanied by other malformations of the genitourinary tract, such as hypospadias, or may occur in the context of inherited disorders such as Prader-Willi syndrome and Noonan syndrome.

Testicular descent occurs in two phases. During the first transabdominal phase, the testes come to lie within the lower abdomen or brim of the pelvis. This phase is controlled by the hormone Mullerian-inhibiting substance. In the second inguinoscrotal phase, the testes descend through the inguinal canal into the scrotal sac. This phase is androgen-dependent and is thought to be mediated by androgen-induced release of calcitonin gene-related peptide from the genitofemoral nerve. The testes may arrest anywhere along their pathway of descent; the most common site is in the

inguinal canal. Arrest within the abdomen is uncommon, accounting for approximately 5% to 10% of cases. Even though testicular descent is controlled by hormonal factors, cryptorchidism is only rarely associated with a well-defined hormonal disorder.

MORPHOLOGY

Cryptorchidism is usually unilateral, being bilateral in 25% of patients. Cryptorchid testes are small and firm. The histologic changes in the malpositioned testis begin as early as 2 years of age. Early on, **thickening of the basement membrane** of the spermatic tubules is seen (Fig. 21.15). Subsequent loss of spermatogonia leaves the tubules with only Sertoli cells. The scarred tubules may appear as dense cords of hyaline connective tissue associated with a concomitant increase in interstitial stroma. Germ cell neoplasia in situ (described later) may be present. Leydig cells are spared and therefore appear relatively prominent. Similar histologic changes may also be seen in the contralateral (descended) testis in males with unilateral cryptorchidism, suggesting that cryptorchidism is a marker of an intrinsic defect in gonadal development.

Clinical Features

Cryptorchidism is asymptomatic and comes to attention when the scrotal sac is discovered to be empty by the patient, a parent, or a physician. In addition to sterility, cryptorchidism may be associated with other morbidity such as crushing injuries due to trauma to the inguinal region and inguinal hernia (10% to 20% of cases).

During the first year of life, most inguinal cryptorchid testes descend spontaneously into the scrotum. If it remains undescended, surgical correction by orchiopexy (placement in the scrotal sac) is required, preferably prior to development of histologic changes. Current recommendations are for hormone injections followed by early orchiopexy to be performed at 6 to 12 months of age if medical treatment fails. Even with this corrective procedure, deficient spermatogenesis has been reported in 10% to 60% of patients, and repositioning does not eliminate the elevated risk of cancer (four- to 10-fold increased risk). Since the contralateral normally descended testis is also at higher risk for malignancy, it is believed that cryptorchidism and the associated risk for germ cell neoplasia is linked to an in utero defect in gonadal cell development (see discussion of testicular dysgenesis syndrome later), rather than the abnormal anatomic position.

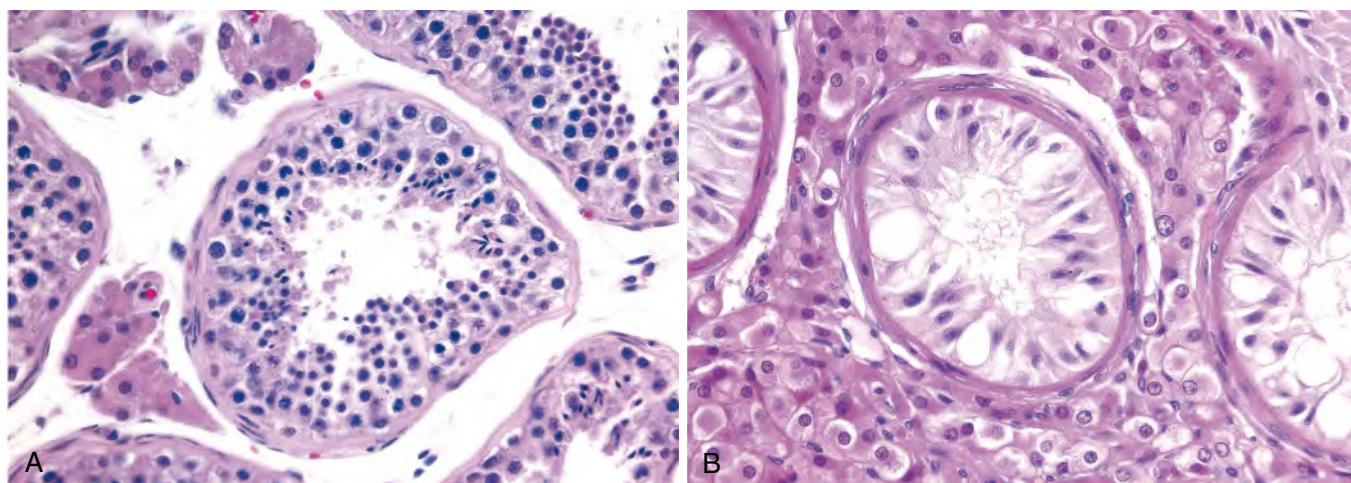


Fig. 21.15 Cryptorchidism. (A) Normal testis shows tubules with active spermatogenesis. (B) Testicular atrophy in cryptorchidism. The tubules contain Sertoli cells but lack spermatogenesis. There is thickening of basement membranes and an apparent increase in interstitial Leydig cells.

KEY CONCEPTS

Cryptorchidism

- Cryptorchidism refers to incomplete descent of the testis from the abdomen to the scrotum and is present in about 1% of 1-year-old male children.
- Bilateral or, in some cases, even unilateral cryptorchidism is associated with tubular atrophy, diminished spermatogenesis, and sterility.
- The cryptorchid testis carries a four- to 10-fold higher risk for testicular cancer.
- Medical therapy and orchiopexy reduces, but does not completely eliminate, the risk of sterility and cancer.

Regressive Changes

Atrophy and Decreased Fertility

Testicular atrophy may be caused by one of several conditions, including (1) progressive atherosclerotic narrowing of the blood supply in old age; (2) end-stage inflammatory orchitis; (3) cryptorchidism; (4) hypopituitarism; (5) generalized malnutrition or cachexia; (6) irradiation; (7) prolonged administration of antiandrogens (e.g., for treatment of carcinoma of the prostate); (8) cirrhosis; (9) primary failure of genetic origin, such as in Klinefelter syndrome; and (10) exhaustion atrophy, which may follow persistent stimulation by high levels of follicle-stimulating pituitary hormone. The gross and microscopic alterations follow the pattern already described for cryptorchidism. In extreme cases there is complete regression (“vanishing testis”).

Inflammation and Infections

Inflammatory disorders are more common in the epididymis than in the testes. Several, including tuberculosis and gonorrhea, arise first in the epididymis and only involve the testis secondarily, while others, such as syphilis and mumps, involve the testis first.

Nonspecific Epididymitis and Orchitis

Epididymitis and possible subsequent orchitis are commonly related to infections in the urinary tract (cystitis, urethritis, prostatitis), which reach the epididymis and the testis through the vas deferens or the lymphatics of the spermatic cord. The cause of epididymitis varies with the age of the patient. Though uncommon in children, epididymitis in childhood is usually associated with a congenital genitourinary abnormality and infection with gram-negative rods. In sexually active males younger than age 35 years, the sexually transmitted pathogens *C. trachomatis* and *Neisseria gonorrhoeae* are most frequent. In males older than age 35, common urinary tract pathogens, such as *E. coli* and *Pseudomonas*, are responsible for most infections.

MORPHOLOGY

As elsewhere in the body, bacterial infection induces acute inflammation characterized by congestion, edema, and infiltration by neutrophils. The infection usually starts in the interstitial connective tissue, but then rapidly extends to involve the tubules, sometimes culminating in abscess formation or suppurative necrosis of the entire epididymis (Fig. 21.16). From here infection often extends to the testis, where it evokes a similar inflammatory reaction. Inflammation of the epididymis and testis is often followed by fibrous scarring, which may produce sterility. Usually, the Leydig cells are not destroyed, and as a result androgen production by the testis is relatively unaffected.

Granulomatous (Autoimmune) Orchitis

Idiopathic granulomatous orchitis, a rare inflammatory disorder, usually presents in middle age as a moderately tender testicular mass of sudden

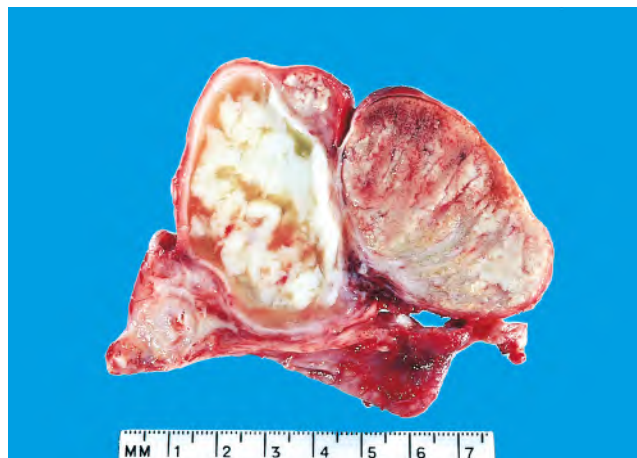


Fig. 21.16 Acute epididymitis caused by gonococcal infection. The epididymis is replaced by an abscess. Normal testis is seen on the right.

onset, sometimes associated with fever. It sometimes appears insidiously, however, as a painless testicular mass mimicking a testicular tumor; hence its importance. Histologically, the orchitis is distinguished by granulomas restricted to spermatic tubules. The process must be distinguished from mycobacterial infection (see later), which often involves the epididymis and produces caseous necrosis. Trauma, infection, extravasated sperm, and autoimmune etiology are postulated as pathogenetic mechanisms.

Infections

Gonorrhea. Extension of infection from the posterior urethra to the prostate, seminal vesicles, and then to the epididymis is a frequent complication of neglected gonococcal infection. In severe cases epididymal abscesses may develop, leading to extensive destruction and scarring. The infection may also spread to the testis and produce suppurative orchitis.

Mumps. Mumps is a systemic viral disease that most commonly affects school-aged children. Testicular involvement is extremely uncommon in this age group. In postpubertal males, however, orchitis occurs in 20% to 30% of cases. Most often, acute interstitial orchitis (marked by infiltrating lymphocytes and plasma cells) develops 1 week after the onset of swelling of the parotid glands. Bilateral involvement can give rise to reductions in fertility.

Tuberculosis. When it involves the male genital tract, tuberculosis almost invariably begins in the epididymis, from where it may spread to the testis. The infection invokes the classic morphologic reactions of caseating granulomatous inflammation characteristic of tuberculosis elsewhere.

Syphilis. The testis and epididymis may be affected in both acquired and congenital syphilis, but almost invariably the testis is involved first. The morphologic pattern of the reaction takes two forms: (1) *obliterative endarteritis* associated with perivascular cuffs of lymphocytes and plasma cells, the histologic hallmark of syphilis, or (2) granulomatous inflammation, a lesion known as a *gumma* (Chapter 8).

Other Infections. Leprosy, sarcoidosis, Crohn's disease, malakoplakia, toxoplasmosis, fungi, other viruses, parasites, and brucellosis are other rare causes of orchitis.

Vascular Disorders

Torsion

Twisting of the spermatic cord typically cuts off the venous drainage of the testis. If untreated, it frequently leads to testicular infarction and thus



Fig. 21.17 Torsion of testis. The dark discoloration is the result of hemorrhage and infarction.

represents one of the few true urologic emergencies. The thick-walled arteries remain patent, producing intense vascular engorgement followed by hemorrhagic infarction.

There are two settings in which testicular torsion occurs. Neonatal torsion occurs either in utero or shortly after birth. It lacks any associated anatomic defect to account for its occurrence. “Adult” torsion is typically seen in adolescence and presents with the sudden onset of testicular pain. It often occurs without any inciting injury. If the testis is manually untwisted within approximately 6 hours of the onset of torsion, the affected testis may be spared an orchiectomy. In adults, torsion results from a bilateral anatomic defect that leads to increased mobility of the testes (*bell-clapper abnormality*).

MORPHOLOGY

Depending on the duration of torsion, the morphologic changes range from intense congestion to widespread hemorrhage and testicular infarction (Fig. 21.17). In advanced stages, the swollen testis consists entirely of soft, necrotic, hemorrhagic tissue.

Spermatic Cord and Paratesticular Tumors

Lipomas are common lesions involving the proximal spermatic cord, identified at the time of inguinal hernia repair. Although diagnosed as “lipomas,” many of these lesions probably represent retroperitoneal adipose tissue that has been pulled into the inguinal canal along with the hernia sac, rather than a true neoplasm.

The most common benign paratesticular tumor is *adenomatoid tumor*. Although these lesions are mesothelial in nature, they are not referred to as mesotheliomas to distinguish them from other mesothelial lesions that may occur at this site. Adenomatoid tumors are usually small nodules (<5 cm), typically occurring near the upper pole of the epididymis. The importance of this lesion is that it is one of the few benign tumors that occur near the testis. If the pathologist can identify the nature of this lesion in intraoperative frozen sections, local excision of the adenomatoid tumor can spare the patient from an orchiectomy.

The most common malignant paratesticular tumors are rhabdomyosarcoma in children and liposarcoma in adults.

Testicular Tumors

Testicular neoplasms span an amazing gamut of histologic types. The 2022 WHO classification (Table 21.6) lists numerous categories, two of which are responsible for most lesions, germ cell tumors (GCTs) (95%) and sex cord–stromal tumors, both of which are discussed here.

Germ Cell Tumors

GCTs predominantly affect males of European descent between 15 and 45 years of age and are the most common cancer in this age group. GCTs are increasing in incidence and occur at variable rates across populations, both for unclear reasons. The lifetime risk is highest in Northern Europe and New Zealand (including the Māori population) and is lowest in Africa and Asia. In the United States in 2024, it is predicted that there will be 9760 new cases diagnosed and 500 deaths from GCTs.

Pathogenesis

Both environmental exposures and inherited and acquired genetic abnormalities contribute to the development of germ cell neoplasia. In the next section, we touch on important environmental and genetic factors and how their interplay appears to lead to the stepwise development of GCTs (Fig. 21.18).

Environmental Factors. The role of environmental factors is inferred in part from population migration studies. For example, the incidence of testicular GCTs in Finland is about two times lower than in Sweden, but second-generation Finnish immigrants to Sweden have a tumor incidence that approaches that of the Swedish population. Testicular GCTs are also associated with a spectrum of disorders collectively known as *testicular dysgenesis syndrome*. Components of this syndrome include cryptorchidism, hypospadias, and poor sperm quality. It has been proposed that these conditions are increased by in utero exposures to pesticides and nonsteroidal estrogens. The most important association is with cryptorchidism, which is seen with approximately 10% of testicular GCTs. Curiously, Klinefelter syndrome is associated with a greatly

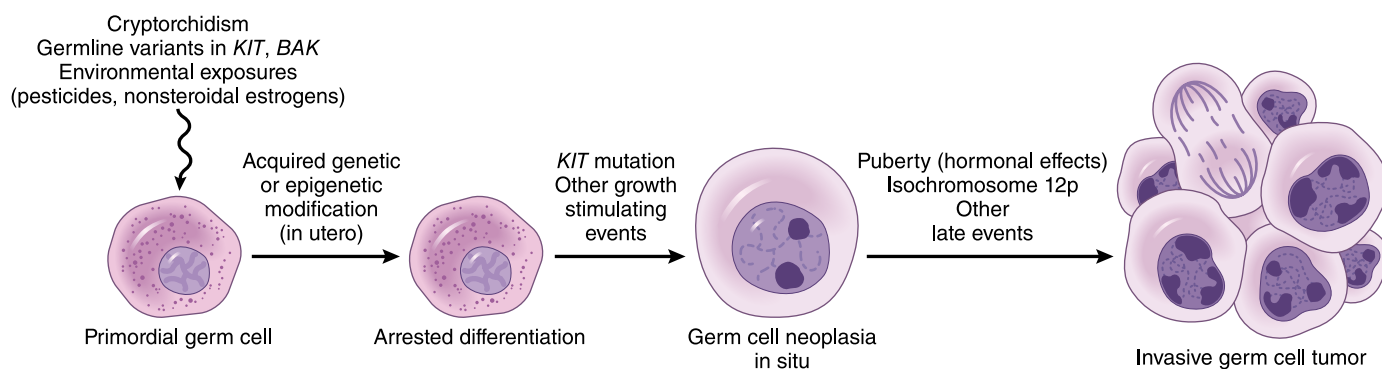


Fig. 21.18 Salient environmental and genetic risk factors and acquired genetic and epigenetic alterations leading to germ cell tumor development and progression. See text for details.

increased risk (50 times normal) for development of mediastinal GCTs, but these patients do not develop testicular tumors.

Genetic Factors. The relative risk of GCTs is four times higher than normal in fathers and sons of affected patients and is 8 to 10 times higher in brothers. Several genetic loci have been linked to familial GCT risk, including variants in genes encoding the ligand for the receptor tyrosine kinase KIT and encoding BAK, which you will recall is an important inducer of apoptotic cell death (Chapter 1). Interestingly, these genes are also thought to play a role in gonadal development. Associations with variants in genes involved in sex hormone metabolism have also been found.

Steps in Germ Cell Tumor Development. A preponderance of evidence points to a primordial germ cell with an acquired defect in differentiation as the cell of origin of GCTs. The next step consists of activation of growth factor receptor signaling, often by activating mutations in the KIT receptor tyrosine kinase, which appear to give rise to a precursor lesion in transformed embryonic germ cells arrested in the gonocyte stage termed *germ cell neoplasia in situ*. This precursor lesion is found in about 90% of testes involved by germ cell neoplasms and is associated with all types of GCTs except spermatocytic tumor and unusual types that arise in infancy (Fig. 21.19). Germ cell neoplasia in situ is also frequently found in testes at high risk for developing GCTs, such as cryptorchid testes. Germ cell neoplasia in situ is believed to arise in utero and stay dormant until puberty, when hormonal influences may stimulate germ cell growth. The lesional cells retain the expression of the transcription factors OCT3/4 and NANOG, which are important in maintenance of pluripotent stem cells. Progression to full-blown GCTs is strongly associated with reduplication of the short arm of chromosome 12 (isochromosome 12p), a cytogenetic alteration that is invariably found in invasive GCTs regardless of histologic type. There are several candidate cancer genes present in this region, including *KITLG* (which encodes KIT ligand), *NANOG*, *KRAS*, *BCAT1*, and *CCND2*, although none are specifically implicated. Progression is very likely, if not inevitable, as about 70% of individuals with germ cell neoplasia in situ (e.g., in cryptorchid testes) develop invasive GCTs within 5 to 7 years of diagnosis.

Classification. A simple classification of the most common types of testicular tumors is presented in Table 21.6. Two broad groups are

TABLE 21.6 Pathologic Classification of Common Testicular Tumors

Germ Cell Tumors Derived From Germ Cell Neoplasia In Situ
Noninvasive germ cell neoplasia
Germ cell neoplasia in situ
Gonadoblastoma
Tumors of a single histologic type (pure forms)
Seminoma
Nonseminomatous germ cell tumors
Embryonal carcinoma
Yolk sac tumor, postpubertal type
Choriocarcinoma
Teratoma, postpubertal type
Teratoma with somatic-type malignancy
Nonseminomatous germ cell tumors of more than one histologic type
Mixed germ cell tumor
Germ Cell Tumors Unrelated to Germ Cell Neoplasia In Situ
Spermatocytic tumor
Teratoma, prepubertal type
Yolk sac tumor, prepubertal type
Mixed teratoma and yolk sac tumor, prepubertal type
Sex Cord–Stromal Tumors
Pure tumors
Leydig cell tumor
Sertoli cell tumor
Other

From Moch H, Amin MB, Berney DM, et al. The 2022 World Health Organization Classification of Tumours of the Urinary System and Male Genital Organs-Part A: Renal, Penile, and Testicular Tumours. *Eur Urol.* 2022;82(5):458-468.

recognized. *Seminomatous tumors* are composed of cells that resemble primordial germ cells or early gonocytes. *Nonseminomatous tumors* may be composed of undifferentiated cells that resemble embryonic stem cells, as in the case of embryonal carcinoma, but the malignant cells may also differentiate along other lineages, generating yolk sac tumors, choriocarcinomas, and teratomas. GCTs may be composed of

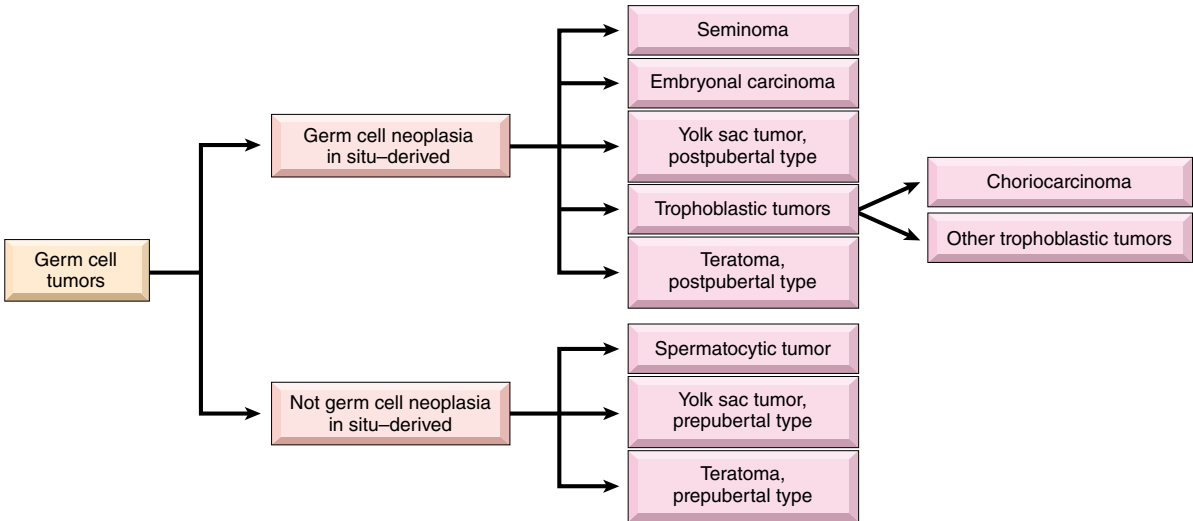


Fig. 21.19 Schematic representation of the ontogeny of germ cell tumors and the two pathways (one associated with germ cell neoplasia in situ, the other not associated). (Modified from Berney DM, Looijenga LH, Idrees M, et al. Germ cell neoplasia in situ [GCNIS]: evolution of the current nomenclature for testicular preinvasive germ cell malignancy. *Histopathology.* 2016;69:7-10.)

a single cell type (pure) or mixtures of seminomatous and/or non-seminomatous components (referred to as mixed germ cell tumors when composed of tumors with more than one GCT component).

Seminoma. Seminoma is the most common type of GCT, making up about 50% of these tumors overall. The peak incidence is in the fourth decade. An identical tumor arises in the ovary, where it is called *dysgerminoma* (Chapter 22), and in the central nervous system, usually in midline structures such as the pineal gland, where it is referred to as *germinoma*.

MORPHOLOGY

Seminomas produce bulky masses, sometimes 10 times the size of the normal testis. The typical seminoma has a homogeneous, gray-white, lobulated cut surface, usually devoid of hemorrhage or necrosis (Fig. 21.20). Generally, the tunica albuginea is not penetrated, but occasionally extension to the epididymis, spermatic cord, or scrotal sac occurs.

Seminomas frequently occur in a pure form but may also be a component of mixed germ cell tumors. The lesion is composed of sheets of uniform cells divided into poorly demarcated lobules by delicate fibrous septa containing a lymphocytic infiltrate (Fig. 21.21A); in some tumors, ill-defined granulomas also are present, presumably as part of a host response to the neoplasm. **The classic seminoma cell is round to polyhedral in shape with a distinct cell membrane; clear or watery-appearing cytoplasm; and a large, central nucleus with one or two prominent nucleoli** (Fig. 21.21B). The cytoplasm

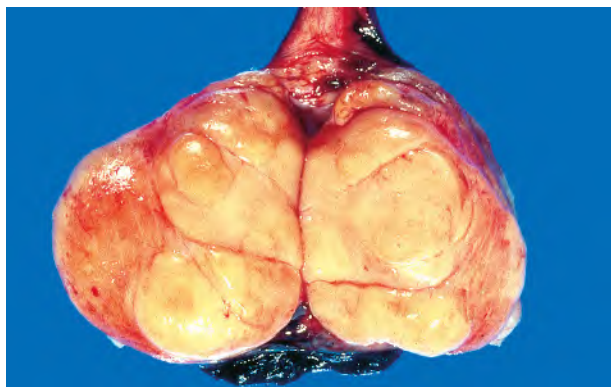


Fig. 21.20 Seminoma of the testis, appearing as a well-circumscribed, pale, fleshy, homogeneous mass on cut surface.

contains varying amounts of glycogen. By immunohistochemistry, the tumor cells are typically positive for KIT, OCT3/4, and podoplanin and negative for cytokeratin. Approximately 15% of seminomas contain syncytiotrophoblasts. In this subset of patients, serum human chorionic gonadotropin (hCG) levels are elevated, though not to the extent seen in patients with choriocarcinoma.

Spermatocytic Tumor. Spermatocytic tumors are uncommon, representing less than 1% of all testicular germ cell neoplasms. In contrast to other GCTs, affected individuals are generally older (usually more than 65 years old). This slow-growing tumor does not metastasize, and when treated by surgical resection it has an excellent prognosis. Although it bears some morphologic resemblance to seminoma, the origin and pathogenesis of spermatocytic tumor are quite distinct, as the tumor is not associated with germ cell neoplasia in situ, lacks isochromosome 12p, and instead is characteristically associated with gain of chromosome 9q (including amplification of the short arm of chromosome 9 including *DMRT1*, a gene involved in sex determination and gonadal development.)

MORPHOLOGY

Grossly, spermatocytic tumor tends to be circumscribed and fleshy and is occasionally myxoid, sometimes with cystic regions. The tumor characteristically contains three intermixed cell populations: (1) medium-sized cells containing a round nucleus, spireme-type chromatin (similar to that seen in the meiotic phase of normal spermatocytes) with occasional nucleoli, and eosinophilic cytoplasm; (2) smaller cells with dense chromatin and with a narrow rim of eosinophilic cytoplasm resembling secondary spermatocytes; and (3) scattered giant cells, either uninucleate or multinucleate. In contrast to seminoma, spermatocytic tumor lacks inflammatory infiltrates and syncytiotrophoblasts, does not occur at extratesticular sites, and is never admixed with other GCTs. Tumors may very rarely have sarcomatous change. They are positive for KIT and SALL4, a transcription factor involved in stem cell maintenance.

Embryonal Carcinoma. The peak incidence for embryonal carcinoma is in the 20- to 30-year-old age group, about a decade earlier than that of seminoma. Embryonal carcinoma is the second most common form of pure GCT, although it is frequently admixed with other germ cell components; it is more aggressive than seminoma.

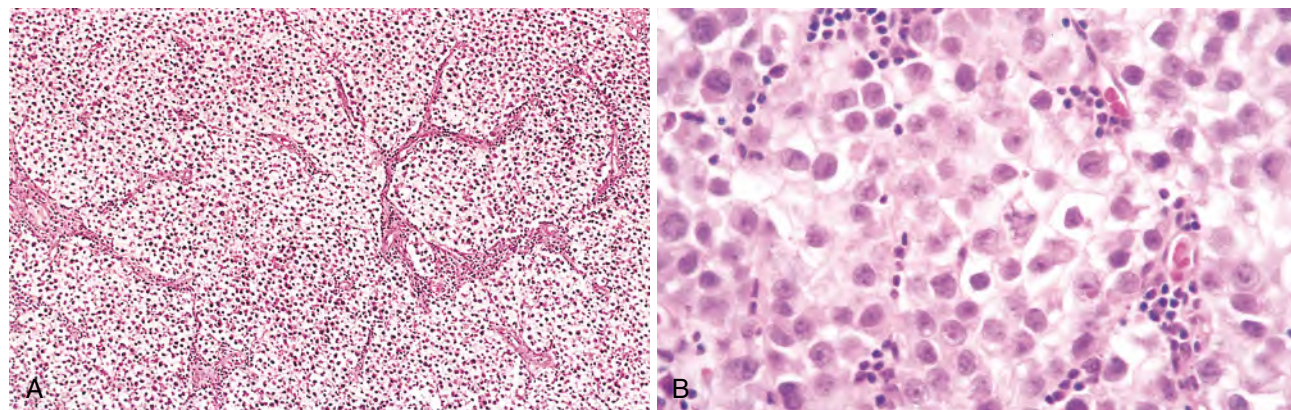


Fig. 21.21 Seminoma. (A) Low magnification shows sheets of pale tumor cells divided into poorly demarcated lobules by delicate septa containing reactive lymphocytes. (B) High magnification reveals large cells with distinct cell borders, pale nuclei, prominent nucleoli, and abundant cytoplasm.

MORPHOLOGY

Embryonal carcinomas tend to be locally aggressive, frequently extending through the tunica albuginea into the epididymis or spermatic cord. On cut surface, the tumor often has a variegated appearance due to the frequent presence of foci of hemorrhage or necrosis (Fig. 21.22). Histologically, the cells grow in alveolar or tubular patterns, sometimes with papillary folds. More undifferentiated lesions may display sheets of cells with cleft-like spaces (Fig. 21.23). The neoplastic cells have an epithelial appearance, are large and anaplastic, and have hyperchromatic nuclei with multiple, prominent nucleoli. The cell borders are usually indistinct, and there is considerable variation in cell and nuclear size and shape (pleomorphism). Mitotic figures and tumor giant cells are frequently seen. Vascular-lymphatic invasion is common. Like seminoma, embryonal carcinoma stains positively for OCT3/4 and SALL4 but differs in that it is typically also positive for cytokeratin and negative for KIT and podoplanin.

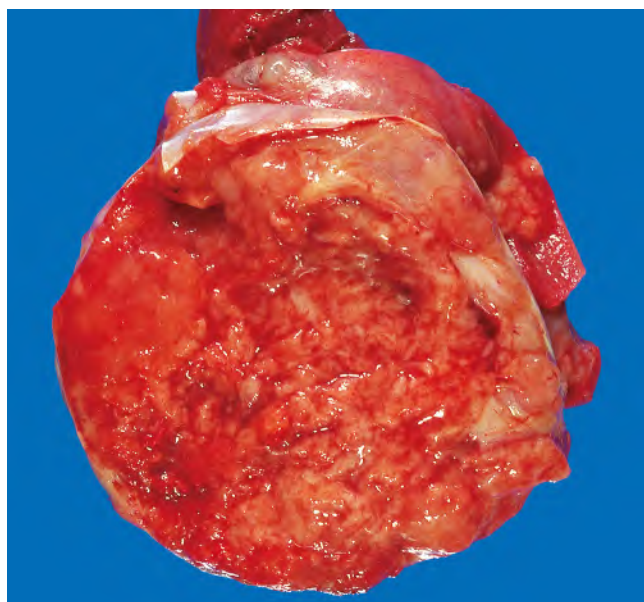


Fig. 21.22 Embryonal carcinoma. In contrast to the seminoma illustrated in Fig. 21.20, as shown here, embryonal carcinoma often produces a hemorrhagic mass.

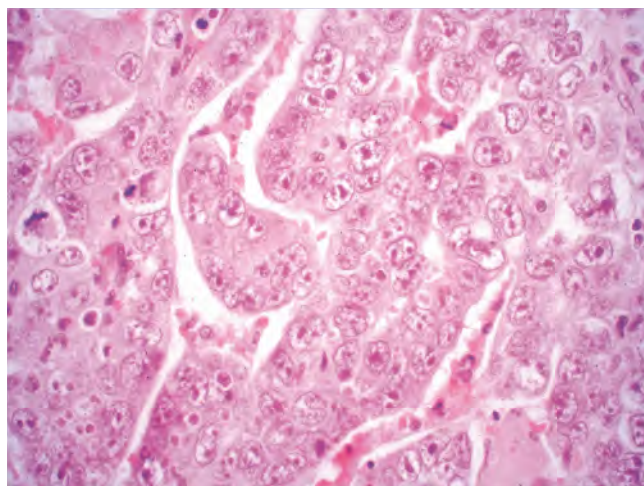


Fig. 21.23 Embryonal carcinoma composed of cells with large, hyperchromatic nuclei arranged in sheets and poorly formed glands.

Yolk Sac Tumor. Also known as *endodermal sinus tumor*, prepubertal yolk sac tumors are the most common testicular tumor in infants and children up to 3 years of age. In this age group, it is not associated with germ cell neoplasia in situ as a precursor lesion, and it has a very good prognosis. By contrast, postpubertal yolk sac tumor is rarely “pure” and more frequently occurs in combination with embryonal carcinoma or other germ cell components.

MORPHOLOGY

Pre- and postpubertal yolk sac tumors have similar histology; the chief difference is that prepubertal tumors are not known to be associated with germ cell neoplasia in situ and lack other germ cell elements. Yolk sac tumors are typically composed of a lace-like (reticular) network of medium-sized cuboidal, flattened, or spindled cells, which generally show less cytologic atypia than embryonal carcinoma. In addition, papillary structures, solid cords of cells, and a multitude of other, less common growth patterns may be found. In approximately 50% of tumors, structures resembling endodermal sinuses (**Schiller-Duval bodies**) may be seen; these consist of a mesodermal core with a central capillary and a visceral and parietal layer of cells resembling primitive glomeruli. Eosinophilic, hyaline-like globules containing α fetoprotein (AFP) and α_1 -antitrypsin are typically seen in yolk sac tumor cells. The expression of AFP, glypican-3, and cytokeratin is characteristic of yolk sac tumor cells and underscores their resemblance to yolk sac;

Choriocarcinoma. Choriocarcinoma is a highly malignant type of GCT. In its pure form, choriocarcinoma is rare, constituting less than 1% of all GCTs. Serum hCG is invariably elevated, often markedly so. Widespread metastasis occurs early and often associated with hemorrhage at sites of involvement.

MORPHOLOGY

Typically, choriocarcinoma primary tumors are small, rarely larger than 5 cm in diameter. Hemorrhage and necrosis are extremely common. The tumor is composed of two intimately juxtaposed cell types, **syncytiotrophoblasts** and **cytotrophoblasts** (Fig. 21.24). Syncytiotrophoblasts are large multinucleated cells with abundant eosinophilic vacuolated cytoplasm containing hCG, which is readily detected by immunohistochemistry. Cytotrophoblasts are more regular and tend to be polygonal, with distinct borders and clear cytoplasm; they grow in cords or sheets and have a single, fairly uniform nucleus. This neoplasm can also arise in the female genital tract (Chapter 22). When pure, the tumor has a poorer prognosis than other germ cell tumors.

Teratoma. The designation *teratoma* refers to GCTs that have various cellular or organoid components reminiscent of the normal derivatives of more than one germ layer. They may occur at any age from infancy to adulthood. Pure teratoma is fairly common in infants and children and is referred to as the “prepubertal type.” Among GCTs, teratomas are second in frequency in infants and children only to yolk sac tumors. In adults, pure teratomas (called postpubertal teratomas) are rare, constituting 2% to 3% of GCTs; most often teratomas are found mixed with other histologies (a phenomenon that is discussed later). Teratomas that occur early in life also differ biologically and clinically from those that occur in adults. Prepubertal teratomas are not associated with germ cell neoplasia in situ or isochromosome 12p and pursue a benign course. Only a small fraction of teratomas occurring in adults share these features, and are generally considered to be malignant.

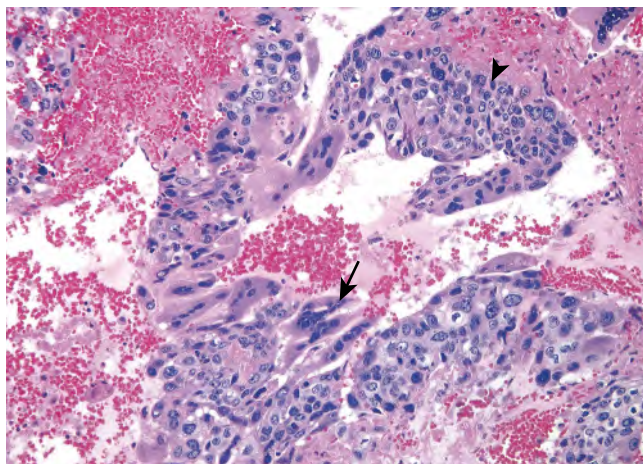


Fig. 21.24 Choriocarcinoma. Both cytotrophoblastic cells (arrowhead) with central nuclei and syncytiotrophoblastic cells (arrow) with multiple dark nuclei embedded in eosinophilic cytoplasm are present. Hemorrhage and necrosis are seen in the upper right field.

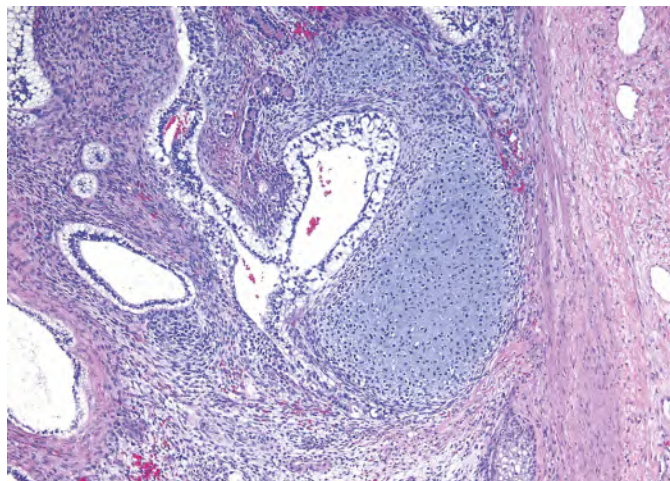


Fig. 21.26 Teratoma of the testis, consisting of a disorganized collection of glands, cartilage, smooth muscle, and immature stroma.

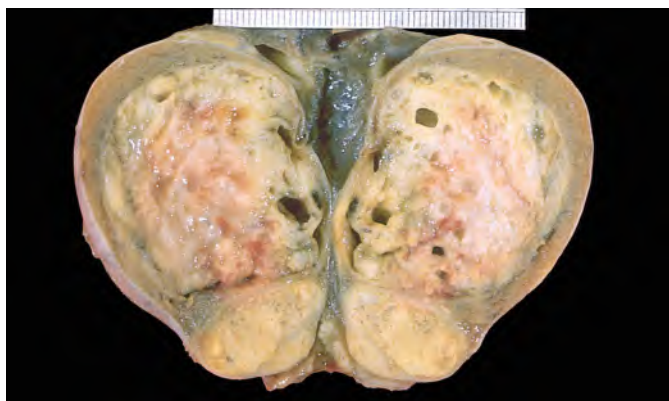


Fig. 21.25 Teratoma of testis. The variegated cut surface with cysts reflects the presence of multiple tissue types.

MORPHOLOGY

Teratomas presenting in adult males are usually large, ranging from 5 to 10 cm in diameter. The presence of a variety of tissues imparts a heterogeneous appearance, with solid, sometimes cartilaginous, and cystic areas (Fig. 21.25). Microscopically, collections of differentiated cells or organoid structures, such as neural tissue, muscle bundles, islands of cartilage, squamous epithelium lining epidermal-like surfaces with or without skin adnexal structures, structures reminiscent of thyroid gland, bronchial epithelium, and bits of intestinal wall or brain substance, are seen embedded in a fibrous or myxoid stroma (Fig. 21.26). Elements may be mature (resembling various adult tissues) or immature (sharing histologic features with fetal or embryonic tissue).

Rarely, malignant non-germ cell (somatic) tumors arise in postpubertal teratomas, a phenomenon referred to as **teratoma with somatic-type malignant transformation**. Transformation may take the form of a squamous cell carcinoma, mucin-secreting adenocarcinoma, sarcoma, or other cancers. The importance of recognizing a somatic-type malignancy arising in a teratoma is that these secondary tumors are chemoresistant; thus, the only hope for cure resides in resection of the primary tumor prior to metastasis. These non-germ cell malignancies retain isochromosome 12p, proving a clonal relationship to the preceding teratoma.

In contrast to their postpubertal counterparts, prepubertal teratomas produce distinct elements that are arranged into structures that more closely resemble normal tissue. Special types include dermoid cysts, which typically contain hair, teeth, and skin. On rare occasions, prepubertal teratoma may be admixed with yolk sac tumor.

Mixed Tumors. About 30% to 40% of GCTs are composed of more than one of the above types. Common mixtures include teratoma, embryonal carcinoma, and yolk sac tumor; seminoma with embryonal carcinoma; and embryonal carcinoma with teratoma.

Clinical Features of Testicular Germ Cell Tumors. Although painless enlargement of the testis is a characteristic feature of GCTs, any solid testicular mass is considered neoplastic until proven otherwise. Biopsy of a testicular neoplasm is associated with a risk of tumor spillage, which would necessitate excision of the scrotal skin in addition to orchiectomy. Consequently, the standard management of a solid testicular mass is radical orchiectomy based on the presumption of malignancy. Occasionally, when a benign lesion is suspected and the lesion can be excised by partial orchiectomy, an intraoperative frozen section examination may be performed in an effort to avoid an unnecessary radical orchiectomy.

Testicular tumors have a characteristic mode of spread. *Lymphatic spread* is common to all forms. In general, retroperitoneal paraaortic nodes are the first to be involved. Subsequent spread may occur to mediastinal and supraclavicular nodes. *Hematogenous spread* is primarily to the lungs, but the liver, brain, and bones may also be involved. Metastases from seminomas typically involve lymph nodes. Hematogenous spread occurs later in the course. Nonseminomatous GCTs tend to metastasize earlier and more frequently via a hematogenous route. The histology of metastases and distant recurrences may differ from that of the testicular lesion. This reflects the fact that GCTs are derived from pluripotent germ cells. Furthermore, a minor component in the primary tumor that is unresponsive to chemotherapy may survive and subsequently become the dominant histologic type in the metastatic site.

Serum Biomarkers of Germ Cell Tumors. GCTs secrete polypeptide hormones and enzymes that can be detected in the patient's peripheral blood. These include hCG, AFP, and lactate dehydrogenase, all of which serve as valuable biomarkers in the diagnosis and management of testicular cancer. The elevation of lactate dehydrogenase correlates with the mass of tumor cells and provides a tool to assess tumor burden.

Marked elevation of serum AFP or hCG levels is produced by yolk sac tumor and syncytiotrophoblastic elements, respectively. Both markers are elevated in more than 80% of individuals with nonseminomatous GCTs at the time of diagnosis. As stated earlier, approximately 15% of seminomas have syncytiotrophoblastic giant cells and minimal elevation of hCG levels. In the context of testicular tumors, serum markers are valuable in several different ways:

- In the initial evaluation of testicular masses
- In the staging of testicular GCTs; for example, after orchiectomy, persistent elevation of hCG or AFP concentrations indicates metastatic spread
- In assessing tumor burden (serum markers are incorporated into the TNM stage to provide prognostic stage grouping)
- In monitoring the response to therapy

The therapy and prognosis of testicular tumors depend largely on the clinical and pathologic stage and the histologic type. Seminoma, which is radiosensitive and chemosensitive, has the best prognosis. More than 95% of patients with stage I and II seminoma are cured by orchiectomy with or without chemotherapy or radiotherapy. In nonseminomatous GCT patients, approximately 90% achieve complete remission with aggressive chemotherapy, and most can be cured. Pure embryonal carcinoma behaves more aggressively than mixed GCTs. Pure choriocarcinoma and mixed GCT with predominantly choriocarcinoma have a poor prognosis.

Tumors of Sex Cord–Gonadal Stroma

As indicated in Table 21.7, sex cord–gonadal stromal tumors are subclassified based on their presumed histogenesis and differentiation. The two most important members of this group, Leydig cell tumors and Sertoli cell tumors, are described here. Pathologic features correlated with malignancy in sex cord–stromal tumors of the testis include large size (greater than 5 cm), necrosis, infiltrative borders, anaplasia, mitotic activity, vascular-lymphatic invasion, and extratesticular extension.

Leydig Cell Tumors. Tumors of Leydig cells often elaborate androgens and in some cases estrogens and corticosteroids as well. They may arise in children or adults. As with other testicular tumors, the most common presenting feature is testicular swelling, but in some patients, gynecomastia, decreased libido, or precocious puberty brings them to clinical attention. Leydig cell tumor is associated with Klinefelter syndrome, cryptorchidism, and hereditary leiomyomatosis, and renal cell carcinoma syndrome (caused by germline mutations in the metabolic enzyme fumarate hydratase). Approximately 10% of Leydig cell tumors in adults are malignant and can produce metastases.

MORPHOLOGY

Leydig cell tumors form circumscribed nodules, usually less than 5 cm in diameter, that have a distinctive golden brown, homogeneous appearance on cut surfaces. Histologically, the tumor cells resemble their normal counterparts. They are large and have round or polygonal cell outlines, abundant granular eosinophilic cytoplasm, and a round central nucleus. The cytoplasm frequently contains lipid droplets, vacuoles, or lipofuscin pigment and, most characteristically, rod-shaped **crystalloids of Reinke**, which are seen in about 25% of tumors.

Sertoli Cell Tumors. Most Sertoli cell tumors are hormonally silent and present as a testicular mass. They are associated with Carney complex (caused by germline mutations in the gene *PRKARIA*, which encodes a cyclic adenosine monophosphate–dependent protein kinase), Peutz-Jeghers syndrome (Chapter 17), and familial adenomatosis

polyposis syndrome (Chapter 17). These neoplasms usually appear as small, firm, circumscribed nodules with a homogeneous gray-white to yellow cut surface. Histologically, the tumor cells are arranged in distinctive trabeculae that tend to form cord-like structures and tubules. Most Sertoli cell tumors are benign, but approximately 10% pursue a malignant course.

Gonadoblastoma

Gonadoblastomas are a rare in situ premalignant lesion that are classified as a noninvasive germ cell neoplasia. They occur in abnormal and/or dysgenetic gonads and are composed of a mixture of germ cells and immature gonadal stromal elements. In some cases, the germ cell component becomes malignant, giving rise to seminoma.

Testicular Lymphoma

Primary testicular lymphomas are uncommon, accounting for 2% to 5% of testicular neoplasms, but are the most common form of testicular neoplasm in males older than 60 years. In most cases, these are aggressive tumors that are disseminated at the time of detection. In contrast to GCTs, tumors are frequently bilateral and involve the spermatic cord. The most common testicular lymphomas, in decreasing order of frequency, are diffuse large B-cell lymphoma, Burkitt lymphoma, and Epstein-Barr virus–positive extranodal NK/T-cell lymphoma (Chapter 13). Testicular lymphomas have a high propensity for central nervous system involvement, which is a frequent site of recurrence.

KEY CONCEPTS

Testicular Tumors

- Germ cell tumors (GCTs) are by far the most common tumor types, and account for 95% of testicular neoplasms.
- Cryptorchidism, infertility, and prior history of GCT in contralateral testis are risk factors.
- Germ cell neoplasia in situ is a precursor lesion associated with most GCTs. Seventy percent of patients with documented germ cell neoplasia in situ will develop invasive GCTs.
- Seminoma, embryonal carcinoma, yolk sac tumors, choriocarcinoma, and teratoma are the most common types of GCTs. Mixed tumors that contain more than one histologic type account for 30% to 40% of GCTs.
- Clinically, testicular GCTs are divided into two groups: seminomas and nonseminomatous tumors. Seminomas spread mainly to paraaortic nodes and are radiosensitive. Nonseminomatous tumors tend to spread earlier via lymphatic and hematogenous routes.
- AFP, hCG, and lactate dehydrogenase can serve as valuable blood biomarkers in the diagnosis and management of GCTs.
- Non-GCTs include sex cord–gonadal stromal tumors and non-Hodgkin lymphoma, which is the most common testicular tumor in males older than 60 years.

Lesions of Tunica Vaginalis

The tunica vaginalis is a mesothelial-lined surface exterior to the testis that may accumulate serous fluid (*hydrocele*) causing considerable enlargement of the scrotal sac. By transillumination it is usually possible to define the clear, translucent character of the contained fluid. Rarely, malignant mesotheliomas and ovarian epithelial-type tumors can arise from the tunica vaginalis.

Hematocele is a collection of blood in the tunica vaginalis. It is an uncommon condition usually encountered following testicular trauma or torsion, or in individuals with systemic bleeding disorders. *Chylocele* refers to the accumulation of lymph in the tunica and is almost always found in patients with elephantiasis who have widespread, severe lymphatic obstruction caused, for example, by filariasis (Chapter 8).

Spermatocele refers to a small cystic accumulation of semen in dilated efferent ducts or ducts of the rete testis. *Varicocele* is a dilated vein in the spermatic cord. Varicoceles may be asymptomatic but have also been implicated in some males as a contributing factor to infertility. They can be corrected by surgical repair.

PROSTATE

The prostate is a retroperitoneal organ encircling the neck of the bladder and urethra that lacks a distinct capsule. In a normal adult, the prostate weighs approximately 20 g and can be divided into four biologically and anatomically distinct regions, the peripheral, central, transition, and periurethral zones (Fig. 21.27). These zones are at risk for different types of proliferative lesions. For example, most hyperplasias arise in the transition zone, whereas most carcinomas originate in the peripheral zone.

Histologically, the prostate consists of glands separated by abundant fibromuscular stroma. The glands are lined by two layers of cells: a basal layer of low cuboidal epithelium covered by a layer of columnar secretory cells (Fig. 21.28), which often contain small papillary infoldings. Testicular androgens control the growth and survival of prostatic cells, and castration leads to widespread apoptosis of prostatic epithelium and atrophy of the prostate.

Only three pathologic processes affect the prostate gland with sufficient frequency to merit discussion: inflammation, benign prostatic hypertrophy (BPH), and tumors. Of these, BPH is the most common and occurs so often in older males that it can almost be viewed as a “normal” part of aging. Prostatic carcinoma is also extremely common in older males and is an important cause of morbidity and mortality. We begin our discussion with consideration of inflammatory processes.

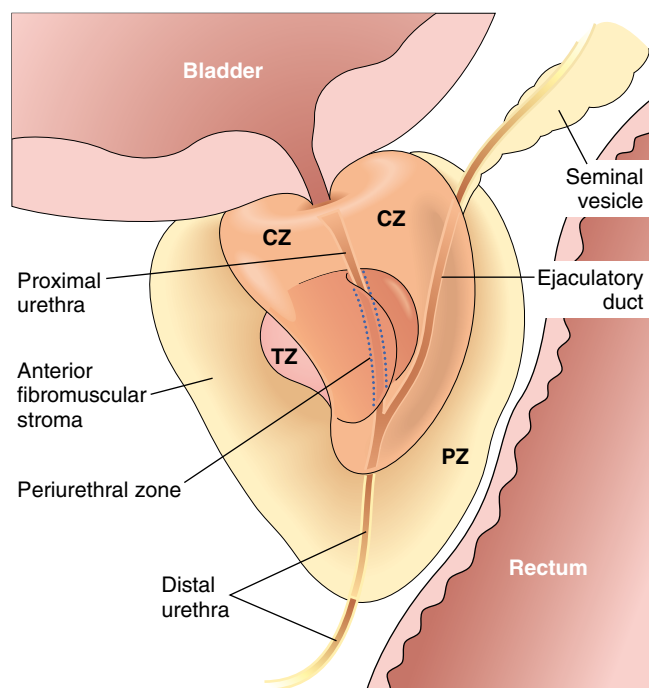


Fig. 21.27 Adult prostate. The normal prostate contains several distinct regions, including a central zone (CZ), a peripheral zone (PZ), a transitional zone (TZ), and a periurethral zone. Most carcinomas arise from the peripheral zone and may be palpable during digital examination of the rectum. Benign prostatic hyperplasia, in contrast, arises from the more centrally situated transitional zone and often produces urinary obstruction.

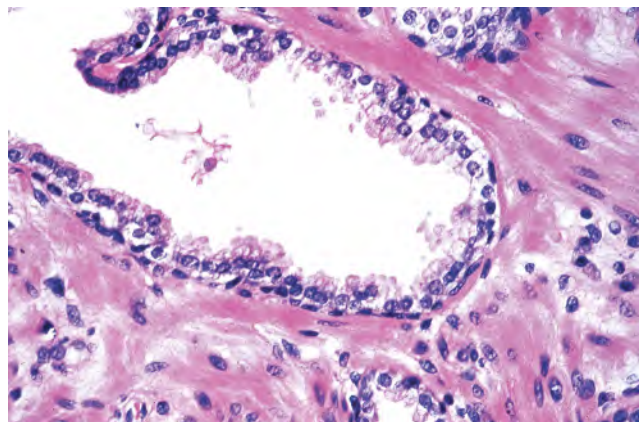


Fig. 21.28 Normal prostate gland, with basal cell and secretory cell layers.

Inflammation

Prostatitis may be divided into several categories, depending on cause, patterns of tissue reaction, and clinical course.

- *Acute bacterial prostatitis* typically results from bacteria similar to those that cause urinary tract infections. Thus, most cases are caused by various strains of *E. coli*, other gram-negative rods, enterococci, and staphylococci. The organisms usually reach the prostate through the reflux of contaminated urine from the posterior urethra or the urinary bladder, but occasionally distant foci of infection seed the prostate through lymphohematogenous routes. Prostatitis sometimes follows surgical manipulation of the urethra or prostate gland (e.g., catheterization, cystoscopy, or urethral dilation). Clinically, acute bacterial prostatitis is associated with fever, chills, and dysuria. On rectal examination the prostate is exquisitely tender and boggy. The diagnosis can be established by urine culture and clinical features. Biopsy of a gland in which acute prostatitis is suspected is contraindicated, as this may lead to sepsis.
- *Chronic bacterial prostatitis* usually presents with low back pain, dysuria, and perineal and suprapubic discomfort, though it may also be asymptomatic. Patients often have a history of recurrent urinary tract infections (cystitis, urethritis) caused by the same organism. Because most antibiotics penetrate the prostate poorly, bacteria find haven in the parenchyma and chronically seed the urinary tract. Diagnosis of chronic bacterial prostatitis depends on the demonstration of leukocytes in expressed prostatic secretions and positive bacterial cultures. The implicated organisms are the same as those that cause acute prostatitis.
- *Chronic abacterial prostatitis*, the most common form of prostatitis, is often referred to as *chronic pelvic pain syndrome*. It is indistinguishable from chronic bacterial prostatitis in terms of signs and symptoms, but there is no history of recurrent urinary tract infection. Expressed prostatic secretions contain more than 10 leukocytes per high-power field, but bacterial cultures are uniformly negative.
- *Granulomatous prostatitis* may be caused by a specific infectious agent or noninfectious stimuli. In the United States, the most common cause is instillation of BCG for treatment of bladder cancer. In this setting, the finding of granulomas in the prostate is of no clinical significance and requires no treatment. Fungal granulomatous prostatitis is typically seen in immunocompromised hosts. Nonspecific granulomatous prostatitis is relatively common and represents a reaction to secretions from ruptured prostatic ducts and acini.
- *Other forms of prostatitis*. Adenoviral and IgG4-associated (Chapter 6) prostatitis have also been described.

Benign Enlargement

Benign Prostatic Hyperplasia

BPH (also referred to as nodular hyperplasia) is the most common benign prostatic disease in males older than 50 years. Approximately 30% of American males of European descent in this age group have moderate to severe symptoms of BPH, and histologic evidence of BPH is found in up to 90% of males by age 80 years. It is not a premalignant lesion.

Etiology and Pathogenesis

Dihydrotestosterone (DHT) is the main androgen in the prostate, where it is formed from testosterone through the action of type 2 5 α -reductase (Fig. 21.29A). This enzyme is expressed primarily in stromal cells and is not expressed in prostatic epithelial cells. Type 1 5 α -reductase is another enzyme that mediates DHT production from testosterone in extraprostatic locations (e.g., liver and skin) and provides an additional source of DHT that reaches the prostate through the blood.

DHT binds to and activates androgen receptors (ARs) found in both stromal and epithelial cells of the prostate. It is more potent than testosterone because it has a higher affinity for ARs and forms a more stable complex with them. Binding of DHT stimulates ARs to translocate from the cytoplasm to the nucleus and activate the transcription of androgen-dependent genes, which encode several growth factors and their receptors. Most important among the upregulated factors are members of the fibroblast growth factor (FGF) family and transforming growth factor- β (TGF- β) (Chapter 1). FGFs, produced by stromal cells, are paracrine regulators of androgen-stimulated epithelial growth during embryonic prostatic development, and some of these pathways may be “reawakened” in adulthood to produce prostatic growth in BPH. TGF- β serves as a mitogen for fibroblasts and other mesenchymal cells but inhibits

epithelial proliferation. Although the ultimate cause of BPH is unknown, it is believed that DHT-induced growth factors act by increasing the proliferation of stromal cells and decreasing the death of epithelial cells.

While it is recognized that androgens play a permissive role in BPH pathogenesis, multiple lines of evidence support a role for estrogens as well. Two different forms of estrogen receptor (ER), namely ER α and ER β , have opposing proliferative and antiproliferative effects on prostate cells, respectively. Effects of estrogens on the prostate are associated with multiple mechanisms including aromatase expression, and paracrine regulation via prostaglandin E $_2$. Estrogens thus contribute to BPH pathogenesis by tipping the balance toward proliferation (Fig. 21.29B).

MORPHOLOGY

In BPH, the weight of the enlarged prostate often increases three- to five-fold (60 to 100 g), and even greater enlargement may be seen. BPH affects the transition zone and thus may encroach on the urethra, compressing it to a slit-like orifice (Fig. 21.30A). On whole mount histology, hyperplastic nodules are seen that vary in color and consistency depending on their cellular content (Fig. 21.30B). Nodules that contain mostly glands are yellow-pink and soft and exude a milky white prostatic fluid. Nodules composed primarily of fibromuscular stroma are pale gray and firm.

Microscopically, individual nodules contain small to large to cystically dilated glands that are separated by bland spindle-shaped stromal cells. The glands are lined by two layers of cells, an inner columnar secretory cell layer and an outer layer of cuboidal or flattened basal epithelium (Fig. 21.30C). Infoldings of the secretory cells may produce a papillary architecture. In markedly enlarged glands, compromise of the vascular supply may produce prostatic infarcts, which may have adjacent areas of squamous metaplasia.

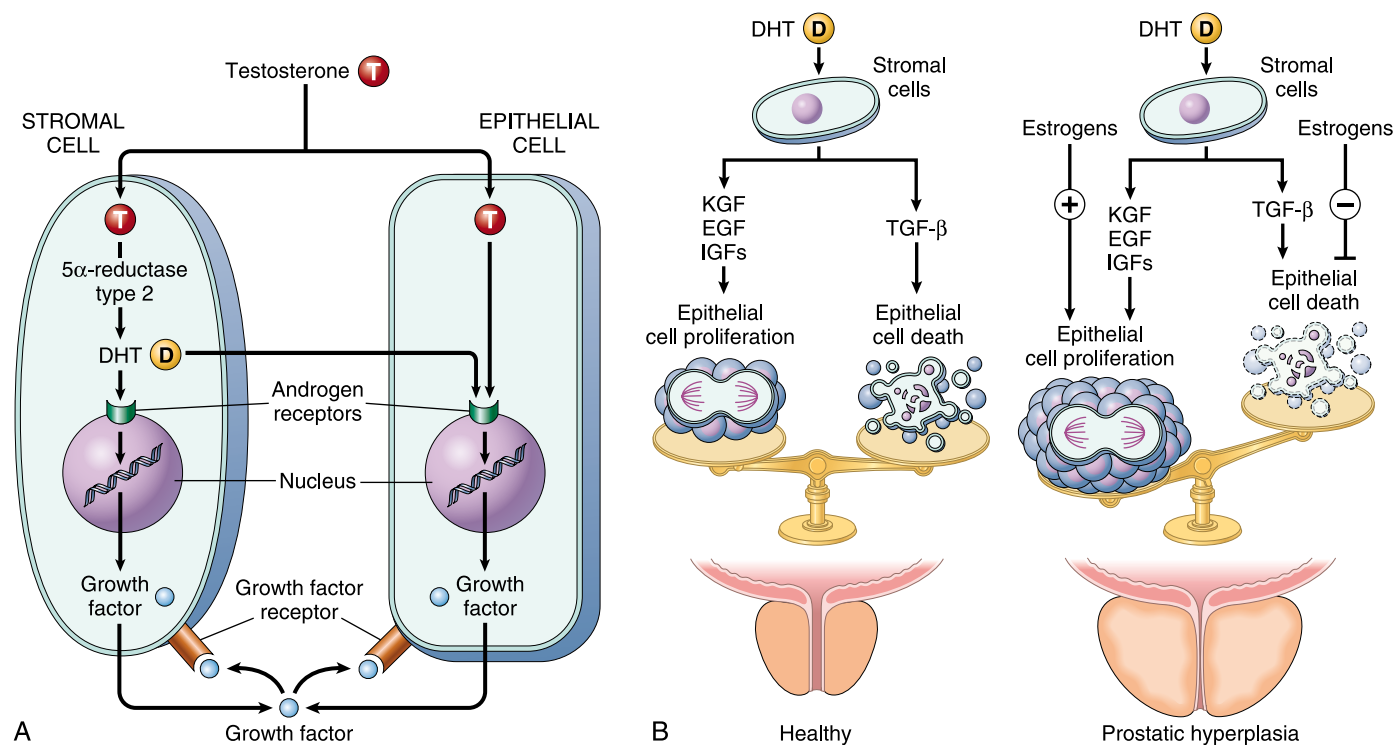


Fig. 21.29 Simplified scheme of the pathogenesis of prostatic hyperplasia. (A) The central role of the stromal cells in generating dihydrotestosterone (DHT) is depicted. DHT may also be produced in skin and liver by both type 1 and type 2 5 α -reductase. (B) The contribution of estrogen in tipping the balance of cell proliferation and cell death toward the former is illustrated. *EGF*, Epidermal growth factor; *IGFs*, insulin-like growth factors; *KGF*, keratinocyte growth factor; *TGF β* , transforming growth factor- β .

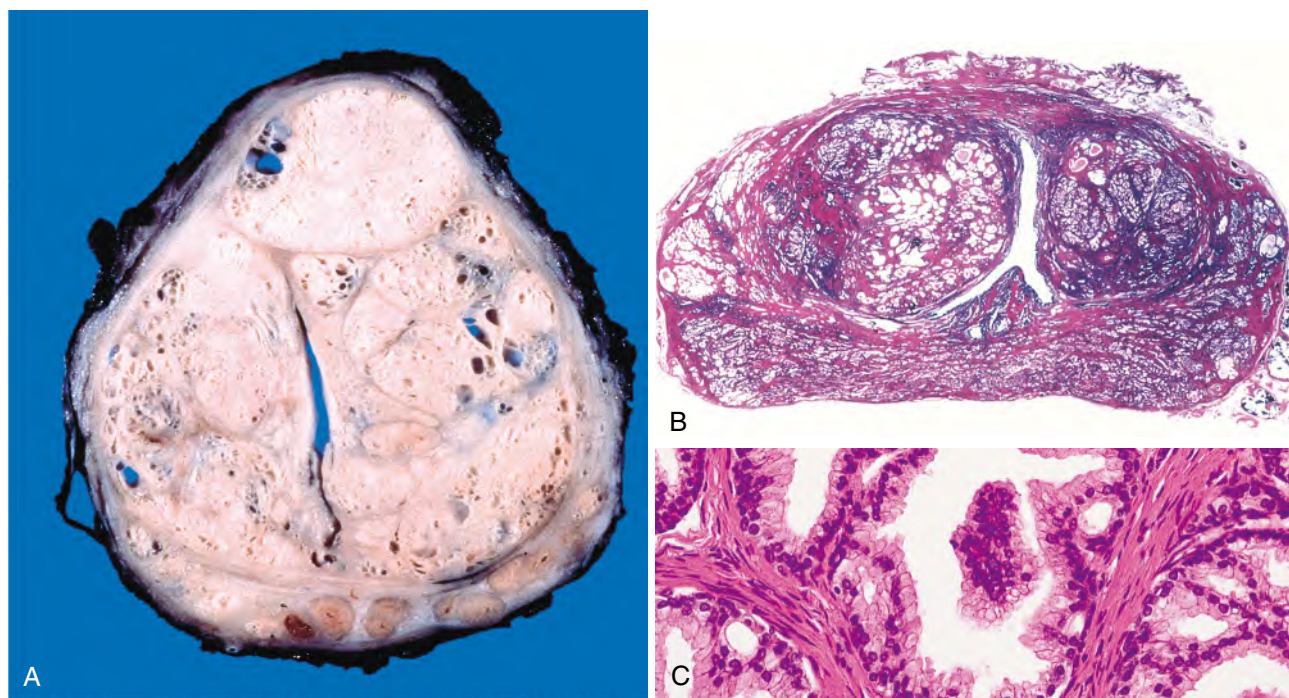


Fig. 21.30 Benign prostatic hyperplasia (BPH). (A) Well-defined nodules of BPH compress the urethra into a slit-like lumen. (B) Microscopic view of a whole mount of the prostate shows nodules of hyperplastic glands on both sides of the urethra. (C) Under high power a nodule composed of hyperplastic glands with papillary epithelial infoldings is seen.

Clinical Features

The main symptoms of BPH are due to urinary obstruction caused by prostatic enlargement and stromal smooth muscle-mediated contraction. The increased resistance to urinary outflow leads to bladder muscular wall hypertrophy and distention, accompanied by urine retention. The inability to empty the bladder completely creates a reservoir of residual urine that is a common site of infection. Patients experience increased urinary frequency, nocturia, difficulty in starting and stopping the stream of urine, overflow dribbling, and dysuria (painful micturition) and have an increased risk of developing bacterial infections of the bladder and kidney. In many cases, sudden, acute urinary retention occurs that requires emergency catheterization for relief.

Symptomatic BPH is usually managed medically with α -adrenergic blockers and 5 α -reductase inhibitors. The former decrease prostate smooth muscle tone via inhibition of α_1 -adrenergic receptors, while the latter physically shrink the prostate by decreasing DHT synthesis. For moderate to severe cases resistant to medical therapy, a wide range of procedures exist. Transurethral resection of the prostate (TURP) was for long the gold standard, but alternative methods that destroy excessive prostatic tissue have been developed with lower morbidity and lower costs. These include high-intensity focused ultrasound (HIFU), laser therapy, hyperthermia, transurethral electrovaporization, and radiofrequency ablation.

KEY CONCEPTS

Benign Prostatic Hyperplasia

- BPH is characterized by proliferation of benign stromal and glandular elements. DHT, an androgen derived from testosterone, is the major hormonal stimulus for proliferation.
- BPH most commonly affects the inner periurethral zone and transition zone of the prostate, producing nodules that compress the prostatic urethra.

- Clinical symptoms and signs are related to urinary obstruction that also predisposes to recurrent urinary tract infections.
- Medical management is based on α -adrenergic blockers and 5 α -reductase inhibitors, which decrease prostatic smooth muscle tone and inhibit DHT production, respectively.

Neoplasms Adenocarcinoma

In the United States, adenocarcinoma of the prostate is the most common form of cancer in males, with an expected 299,010 new cases in 2024, accounting for 29% of all male cancers. Prostate cancer is the second cause of cancer-related death in males, surpassed only by lung cancer. The biologic behavior of prostate cancer spans the gamut from clinically insignificant cancers detected only by microscopic examination to aggressive tumors that are rapidly fatal. It is largely a disease of aging. Based on autopsy studies, the incidence of prostate cancer increases from 20% in males in their 50s to approximately 70% in males between the ages of 70 and 80 years.

Epidemiology and Pathogenesis

An interplay of both environmental exposures and inherited genetic factors contributes to striking differences in the incidence of prostate cancer across geographic locales (Chapter 7). For example, the incidence of prostate cancer in individuals of Japanese descent living in the United States is substantially higher than in Japanese living in Japan, but is only about 50% of the reported incidence in Americans of African descent. As explained below, some genetic factors appear to increase risk due to intrinsic effects on prostatic epithelium, while others may act by modifying the risk associated with environmental exposures. Furthermore, as might be expected of an androgen-sensitive organ, androgens and androgen receptor (AR) function have

central roles in the development, progression, and treatment of prostate cancer.

Environmental Factors. It is hypothesized that exposure to carcinogens, estrogens, and oxidants injures prostatic epithelium, setting the stage for acquisition of genetic and epigenetic changes that lead to cancer development (Fig. 21.31). Broadly speaking, the “Western diet” is a suspect, given the relatively high incidence of prostate cancer in the United States, South America, Western Europe, and Australia. One dietary component that epidemiologic and animal model studies suggest contributes to risk is consumption of charred red meats and animal fats, which leads to formation of carcinogenic heterocyclic aromatic amines and polycyclic aromatic hydrocarbons. These and

other dietary exposures may create oxidant stress in prostatic epithelium, leading to cell injury and inflammation. One piece of evidence linking these exposures to prostate cancer development is the observation that polymorphisms in glutathione-S-transferase (GSTP1), an enzyme involved in detoxification of polycyclic aromatic hydrocarbons, are linked to prostate cancer risk.

Inherited Genetic Factors. Studies of twins and families support the existence of genetic predisposing factors. Males with first-degree relatives affected by the disease have a two-fold increased risk, and numerous germline variants have been identified that are associated with risk. Among the relatively common variants conveying a modest increase in risk are variants in regulatory regions that influence the expression of the oncogene *MYC*. Other rare variants have been identified that are linked to a high risk of early-onset, aggressive disease. Among these are mutations that disrupt the function of several DNA repair genes, including loss-of-function mutations in *BRCA2* (required for repair of DNA breaks by homologous recombination) and in DNA mismatch repair genes (as part of Lynch syndrome; see Chapter 17). A variant in the gene encoding the transcription factor *HOXB13*, which plays a role in prostatic development during embryogenesis, also has been associated with a severalfold increase in risk.

Androgens. Like their normal counterparts, the growth and survival of prostate cancer cells depend on androgens, which bind to the androgen receptor (AR) and induce the expression of pro-growth and pro-survival genes. The importance of androgens in maintaining the growth and survival of prostate cancer cells can be seen in the therapeutic effect of castration or treatment with antiandrogens, which usually induce disease regression. Unfortunately, most tumors eventually become resistant to androgen blockade. Tumors escape through a variety of mechanisms, including acquisition of hypersensitivity to low levels of androgen (e.g., through AR gene amplification); ligand-independent AR activation (e.g., via splice variants that lack the ligand binding domain); mutations in AR that allow it to be activated by nonandrogen ligands; and other mutations or epigenetic changes that activate alternative signaling pathways and bypass the need for AR altogether.

Acquired Genetic and Epigenetic Alterations. As with other cancers, prostate cancer is caused by genetic and epigenetic alterations that altered the expression of tumor suppressor genes and oncogenes, ultimately leading to acquisition of the hallmarks of cancer (Chapter 7). **The most common genetic alteration in prostate cancer is a chromosomal rearrangement that juxtaposes the coding sequence of an ETS family transcription factor gene (most commonly *ERG* or *ETV1*) next to the androgen-regulated *TMPRSS2* promoter.** The rearrangements, which occur in approximately half of prostate cancers, lead to androgen-dependent overexpression of ETS. Silencing of the gene encoding p27, an inhibitor of cyclin-dependent kinases, is another common event. DNA sequencing has demonstrated many other recurrent genomic rearrangements, primarily deletions and amplifications, that involve other cancer genes in prostate carcinoma. Among these are amplification of *MYC* and deletion of *PTEN*, which lead to accelerated cell growth and may contribute to resistance to antiandrogen therapy. In late-stage disease, loss of *TP53* (by deletion or mutation), and deletions of *RB* are common, as are amplifications of the AR gene.

Epigenetic events that modify gene expression are also common in prostate cancer. Work carried out as part of The Cancer Genome Atlas project identified several distinct subsets of prostate cancers that are defined by differences in DNA methylation. One particularly frequent early event is epigenetic silencing by DNA methylation of the *Glutathione-S-transferase pi 1* (*GSTP1*) gene, which downregulates *GSTP1*

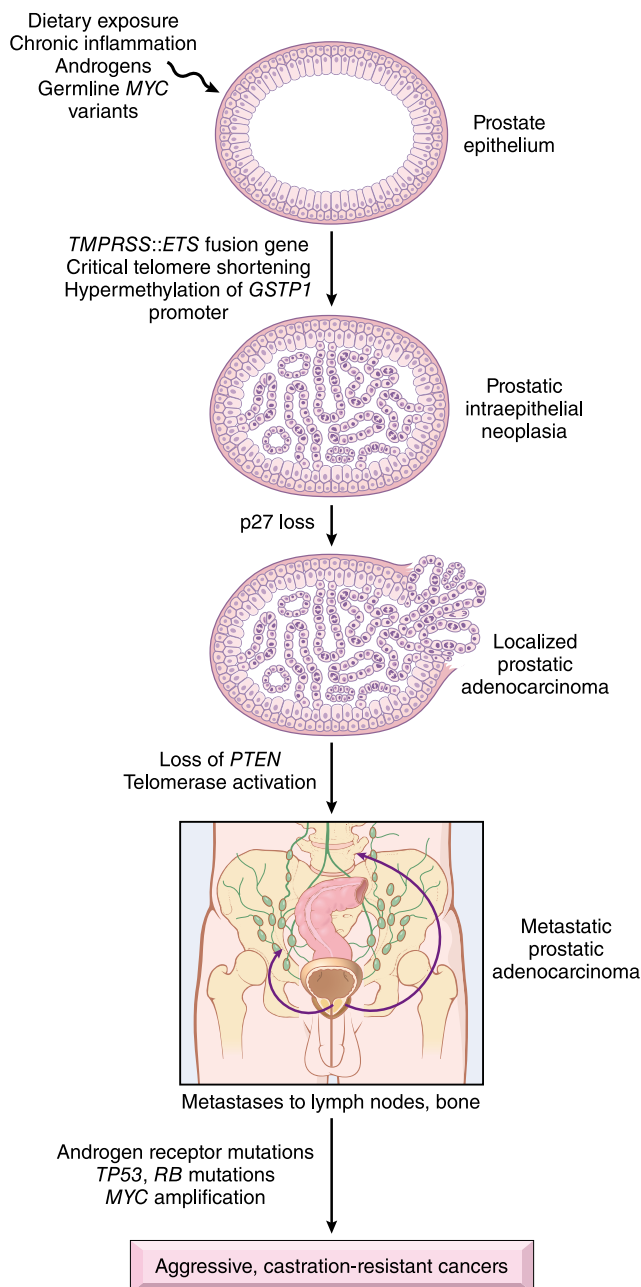


Fig. 21.31 Salient environmental and genetic risk factors and acquired genetic and epigenetic alterations leading to prostate cancer development and progression. See text for details.

expression, possibly enhancing the genotoxic effects of environmental carcinogens. Other genes silenced by epigenetic modifications in a subset of prostate cancers include genes involved in cell cycle regulation (*RB*, *CDKN2A*), maintenance of genomic stability (*MLH1*, *MSH2*), and Wnt signaling (*APC*). The cumulative salient genetic and epigenetic alterations during prostate cancer development and progression are illustrated in Fig. 21.31.

Precursor Lesions. Evidence in favor of stepwise development of prostate cancer (as outlined in Fig. 21.31) includes the existence of a precursor lesion, *high grade prostatic intraepithelial neoplasia (HGPIN)*. A precursor role for HGPIN in at least some cases of prostate cancer is supported by several observations. Both HGPIN and cancer typically predominate in the peripheral zone and are relatively uncommon in other zones, and prostates containing cancer have a higher frequency and a greater extent of HGPIN, which is often seen in proximity to cancer. More directly, many of the molecular changes seen in invasive cancers are also present in HGPIN, strongly supporting the argument that HGPIN is a precursor.

MORPHOLOGY

When the term *prostate cancer* is used without qualification, it refers to the common or acinar variant of prostatic adenocarcinoma. In approximately 70% of cases, carcinoma of the prostate arises in the peripheral zone of the gland, often in a posterior location, where it may be palpable on rectal examination. Characteristically, on cross-section the neoplastic tissue is gritty and firm to palpation, but it is sometimes difficult to visualize by eye (Fig. 21.32). Histologically, most adenocarcinomas consist of glands arranged in well-defined, easily recognized patterns, which are used to grade these tumors (discussed later). The glands are typically smaller than benign glands and are lined by a single uniform layer of cuboidal or low columnar epithelium. In contrast to benign glands, malignant glands have tightly packed cells and characteristically lack branching and papillary infoldings. **The basal cell layer typical of benign glands is absent.** The cytoplasm of the tumor cells ranges from pale-clear to a distinctive amphophilic appearance. Nuclei are enlarged and often contain one or more large nucleoli. There is some variation in nuclear size and shape, but in general pleomorphism is not marked. Mitotic figures are uncommon.

When prostate cancer invades locally, it most commonly involves periprostatic tissue, seminal vesicles, and the base of the urinary bladder, which in advanced disease may produce ureteral obstruction. Metastases spread via lymphatics to

the obturator nodes and eventually to the paraaortic nodes. **Hematogenous spread occurs chiefly to the bones**, particularly the axial skeleton. Bony metastases are typically osteoblastic, a feature that in males points strongly to a prostatic origin (Fig. 21.33). The bones that are most commonly involved, in descending order of frequency, are lumbar spine, proximal femur, pelvis, thoracic spine, and ribs. Tumors may also spread to viscera, but extensive visceral dissemination is the exception rather than the rule.

The diagnosis of prostate cancer on biopsy specimens can be challenging due to several factors. There is often only a scant amount of tissue available for histologic examination in needle biopsies, and malignant glands may be admixed with numerous benign glands (Fig. 21.34). Moreover, the histologic findings may be subtle (leading to underdiagnosis), and there are benign mimickers of cancer that can lead to a misdiagnosis. A few findings are specific, such as perineural invasion (Fig. 21.35), but in general the diagnosis is made based on a constellation of architectural, cytologic, and ancillary findings. As discussed earlier, one distinguishing feature is that benign glands contain basal cells, which are absent in cancer (compare benign normal glands in Fig. 21.34A and benign hyperplastic glands in Fig. 21.30C with cancerous glands in Fig. 21.34B). This distinction can be brought out by using various immunohistologic markers that stain basal cells. Another useful marker is α -methylacyl coenzyme A racemase (AMACR), which is upregulated in prostate cancer. Most prostate cancers are positive for AMACR, the sensitivity varying among studies from 82% to 100%. Such markers, while improving diagnostic accuracy, are still prone to false-positive and false-negative results and must be used in conjunction with the routine hematoxylin and eosin–stained sections.

As already discussed, in approximately 80% of cases, prostatic tissue removed for carcinoma also harbors **high grade prostatic intraepithelial neoplasia (HGPIN)**. HGPIN consists of architecturally benign large, branching prostatic acini lined by atypical cells with prominent nucleoli that may be cytologically identical to carcinoma. Unlike malignant glands, glands involved by HGPIN retain, at least partially, a layer of basal cells and have an intact basement membrane.



Fig. 21.32 Adenocarcinoma of the prostate. An area of cancer is present in the posterior aspect (*lower left*) that has solid gray-white appearance, in contrast to the spongy appearance of benign peripheral zone on the contralateral side.



Fig. 21.33 Metastatic osteoblastic prostatic carcinoma within vertebral bodies.

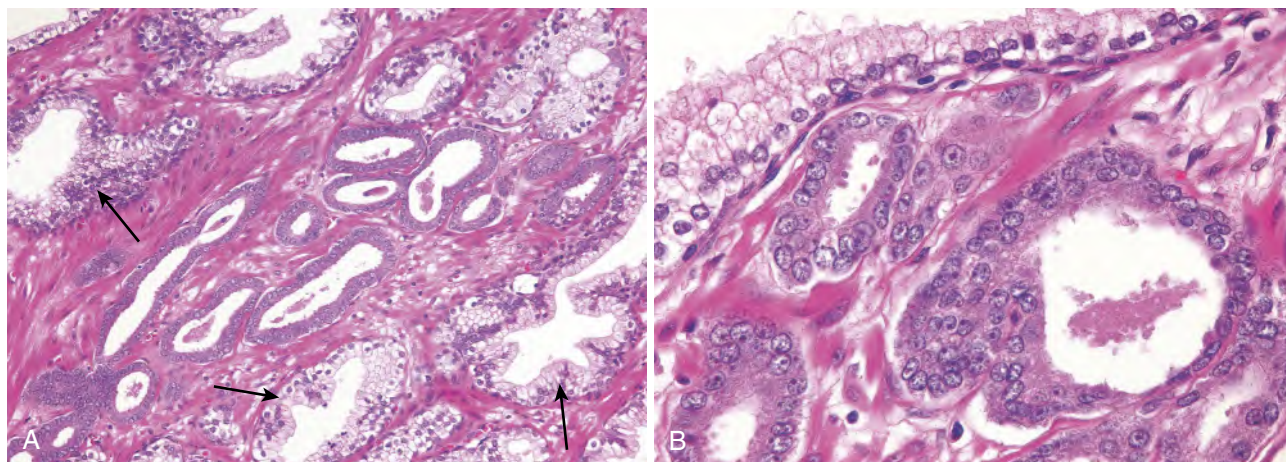


Fig. 21.34 Prostatic adenocarcinoma. (A) Photomicrograph of small focus of adenocarcinoma of the prostate demonstrating small glands crowded in between larger benign glands (arrows). (B) Higher magnification shows several small malignant glands with enlarged nuclei, prominent nucleoli, and dark cytoplasm compared with larger benign gland (top).

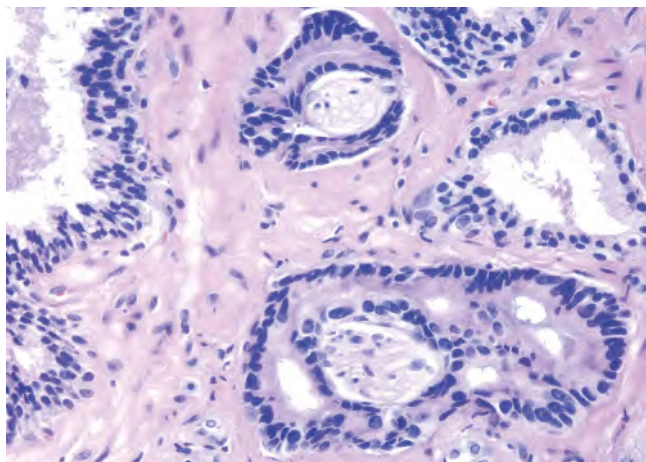


Fig. 21.35 Carcinoma of prostate, showing perineural invasion by malignant glands. Compare to benign gland (left).

Grading and Staging. Grade and stage are the most important prognostic factors in prostate cancer. Grading is performed using the Gleason system, which stratifies prostate cancer into five grades based on glandular patterns of growth. Grade 1 corresponds to well-differentiated tumors in which the neoplastic glands are uniform and round in appearance and are packed into well-circumscribed nodules (Fig. 21.36A). In contrast, in grade 5 tumors, the tumor cells infiltrate the stroma in cords, sheets, and solid nests (Fig. 21.36C). Other grades fall between these extremes (Fig. 21.36B). Most tumors contain more than one pattern; in such instances, a primary grade is assigned to the dominant pattern, and a secondary grade is assigned to the second most frequent pattern. The two numeric grades are then added to obtain a combined Gleason score. Tumors with only one pattern are treated as if their primary and secondary grades are the same, and the number is doubled for the score. Hence, the Gleason score ranges from 2 to 10. Most potentially treatable cancers detected on needle biopsy as a result of screening have Gleason scores of 6 or 7. Tumors with Gleason scores of 8 through 10 tend to be advanced cancers that are less likely to be cured. Although there is some evidence that prostate cancers can become more aggressive with time, most commonly, the Gleason score remains stable over a period of several years.

Presently, Gleason scores are combined into five grade groups (Table 21.7), each with a different prognosis.

Pathologic staging of prostatic cancer is used in combination with the grade to stratify management of prostate cancer. As shown in Table 21.8, pTNM staging is based on tumor extent (T) and presence of nodal or distant metastasis (N and M). Grade is now integrated with TNM pathologic stage data to derive combined prognostic groups that result in “upstaging” of some tumors.

Clinical Features

Localized prostate cancer is asymptomatic and is usually discovered by the detection of a suspicious nodule on rectal examination or elevated serum prostate-specific antigen (PSA) level (discussed later). Digital rectal examination may detect some early prostatic carcinomas because of their posterior location but suffers from both low sensitivity and specificity. Patients with clinically advanced prostatic cancer may present with symptoms of urinary obstruction. Typically, a transrectal needle biopsy is required to confirm the diagnosis.

Measurement of serum PSA levels is widely used to assist with the diagnosis and management of prostate cancer but is controversial. PSA is a product of prostatic epithelium and is normally secreted in semen. It is an androgen-regulated serine protease whose function is to cleave and liquefy the seminal coagulum formed after ejaculation. In normal males, only minute amounts of PSA circulate in the serum. Elevated blood levels of PSA occur in association with both localized and advanced cancer. However, as a screening test for prostate cancer, PSA measurement has suboptimal sensitivity and specificity. PSA is organ specific, but not cancer specific. Although serum levels of PSA are elevated on average to a lesser extent in patients with BPH than in those with prostate cancer, there is considerable overlap in these two groups of patients. Other factors such as prostatitis, prostate gland infarction (e.g., in the setting of BPH), instrumentation of the prostate, and ejaculation also increase serum PSA levels. Of even greater concern, many prostate cancers are so indolent that they are clinically insignificant, and detection of such cancers may lead to overtreatment, with its associated morbidity and economic costs. As a result of these concerns, some countries with national health care systems (e.g., the United Kingdom) have recommended against using PSA as a screening test. By contrast, once the diagnosis of prostate cancer has been established, serial measurement of PSA after treatment has clear-cut value in monitoring recurrence and disease progression.

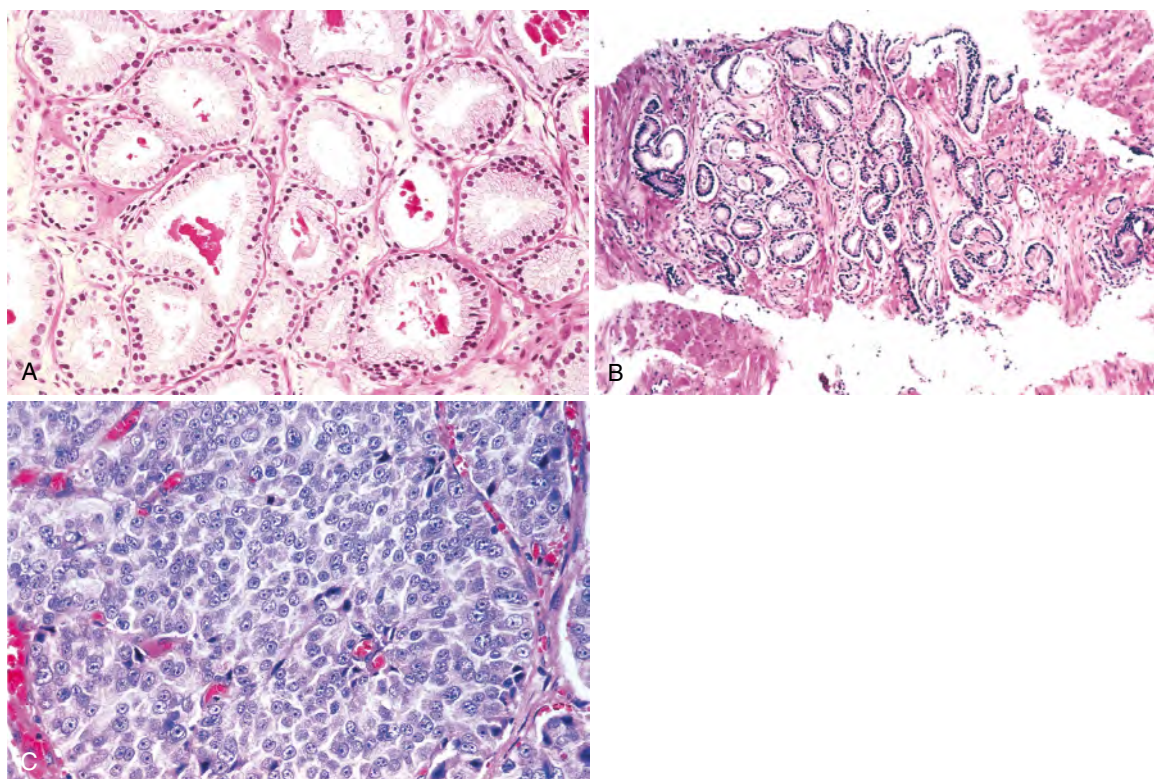


Fig. 21.36 Gleason grading of prostate cancer. (A) Low-grade prostate adenocarcinoma grade group 1 consisting of back-to-back uniform-sized malignant glands lined by a single layer of epithelial cells. (B) Needle biopsy of the prostate with grade group 1 adenocarcinoma (Gleason score 3 + 3 = 6) composed of well-formed, variably sized individual glands. (C) Grade group 5 adenocarcinoma (Gleason score 5 + 5 = 10) composed of sheets of malignant cells with no identifiable gland formation.

TABLE 21.7 Prostate Cancer Gleason Grade Groups

Grade Group 1 (≤ 6)
Only individual discrete well-formed glands
Grade Group 2 (3 + 4)
Predominantly well-formed glands with a lesser component of poorly formed, fused or cribriform glands
Grade Group 3 (4 + 3)
Predominantly poorly formed/fused/cribriform glands with a lesser component of well-formed glands
Grade Group 4 (4 + 4/3 + 5/5 + 3)
Only poorly formed/fused/cribriform glands or predominantly mix of well-formed glands and lack of glands
Grade Group 5 (4 + 5/5 + 4/5 + 5)
Lack gland formation (or with necrosis) with or without poorly formed/fused/cribriform glands

Following the diagnosis of prostate cancer, the next challenge is to determine whether the cancer is likely to progress locally or metastasize. In addition to tumor grade, several genomic tests using gene panels have recently been developed that may enhance the ability to predict which cancers (e.g., low-volume grade group 1) can be followed by active surveillance alone. Patients with localized prostate cancer that is considered to be at high risk for local and systemic spread are treated with surgery or radiation, with or without hormonal manipulation. More than 90% of patients who receive such therapy can expect to live for 15 years. Currently, the most common treatment is radical prostatectomy. The prognosis following radical prostatectomy is based on the pathologic stage, margin status, and Gleason grade. For patients who are not candidates for surgery because of age or other

TABLE 21.8 Pathologic Staging of Prostatic Adenocarcinoma Using the pTNM System: pTNM, AJCC 8th Edition

pTNM Designation	Anatomic Findings
Extent of Primary Tumor (T)	
pT2	Organ confined
pT3	Extraprostatic extension
pT3a	Extraprostatic extension (unilateral or bilateral) or microscopic invasion of bladder neck
pT3b	Tumor invades seminal vesicles
pT4	Tumor is fixed or invades adjacent structures other than seminal vesicles, such as external sphincter, rectum, bladder, levator muscles, or pelvic wall
Definition of Regional Lymph Nodes (N)	
Nx	Regional nodes not accessed
N0	No regional nodal metastases
N1	Metastasis in regional lymph nodes
Definition of Distant Metastases (M)	
M0	No distant metastases
M1	Distant metastases present
M1a	Metastases to distant lymph nodes
M1b	Bone metastases
M1c	Other distant sites

AJCC, American Joint Commission on Cancer.

medical conditions, radiotherapy with either external-beam radiation or placement of radioactive seeds in the prostate (brachytherapy) often provides excellent local control of disease.

Metastatic carcinoma is treated with androgen deprivation therapy, usually achieved by orchiectomy or by administration of synthetic analogs of luteinizing hormone–releasing hormone (LHRH); chronic administration of LHRH agonists desensitizes pituitary cells expressing LHRH receptors, suppressing the release of luteinizing hormone (LH), which is required for testosterone production by Leydig cells. Although androgen deprivation therapy induces remissions, most tumors eventually become resistant to testosterone withdrawal, an event that is a harbinger of disease progression.

KEY CONCEPTS

Carcinoma of the Prostate

- Carcinoma of the prostate is very common in older males. In the United States, it is the most common malignancy in males.
- Prostate carcinomas range from very indolent lesions, which are being increasingly managed by active surveillance, to potentially lethal disease that requires definitive therapy.
- Carcinomas of the prostate arise most commonly in the outer, peripheral gland and may be palpable by rectal examination.
- The most common driver mutations in prostate cancer are gene rearrangements that result in androgen-dependent overexpression of ETS family transcription factors, most commonly ERG or ETV1.
- Grading of prostate cancer by the Gleason system strongly correlates with pathologic stage and is a strong prognosticator. The system is compressed into grade groups that also influence the combined prognostic stage of the tumor.
- Most localized cancers are clinically silent and are detected by routine monitoring of PSA concentrations in older males. Bone metastases, often osteoblastic, typify advanced prostate cancer.
- Serum PSA measurement is a useful but imperfect cancer-screening test, with significant rates of false-negative and false-positive results. Evaluation of PSA concentrations after treatment has great value in monitoring progressive or recurrent disease.

Miscellaneous Tumors

The most common prostate tumors other than conventional prostate adenocarcinoma are other forms of carcinoma. *Ductal adenocarcinoma* arising in peripheral ducts may present in a fashion similar to ordinary prostate cancer, whereas those arising in the larger periurethral ducts may show signs and symptoms similar to urothelial cancer, causing hematuria and urinary obstructive symptoms. Ductal adenocarcinomas are associated with a relatively poor prognosis. Prostate cancers may show squamous differentiation, either following hormone therapy or de novo, resulting in either adenosquamous or pure squamous cancer. Prostate cancers that reveal abundant mucinous secretions in greater than 25% of the tumor are termed *colloid carcinoma of the prostate*.

The most aggressive variant of prostate cancer is *small-cell carcinoma* (also known as neuroendocrine carcinoma). Almost all cases of small-cell carcinoma are rapidly fatal. This cancer is increasing in frequency

and is most often seen as a form of recurrent disease in patients with typical prostate cancer undergoing treatment with antiandrogen therapies. While these therapies control conventional prostate cancers, in some cases they provoke the emergence of an androgen-independent subclone with a neuroendocrine phenotype.

The most common tumor to secondarily involve the prostate is urothelial cancer. Two distinct patterns of involvement exist. Large invasive urothelial cancers can directly invade from the bladder into the prostate. Alternatively, CIS of the bladder can extend into the prostatic urethra and down into the prostatic ducts and acini.

The same mesenchymal tumors described earlier that involve the bladder may also manifest in the prostate. In addition, there exist unique mesenchymal tumors of the prostate derived from the prostatic stroma.

SELECTED READINGS

- Alemamy L, Cubilla A, Halc G et al; HPV VVAP Study Group: role of human papillomavirus in penile carcinomas worldwide, *Eur Urol* 69:953-961, 2016. [Review of the role of HPV in penile carcinoma].
- Berney DM, Looijenga LH, Idrees M, et al: Germ cell neoplasia in situ (GCNIS): evolution of the current nomenclature for testicular pre-invasive germ cell malignancy, *Histopathology* 69:7-10, 2016.
- Cancer Genome Atlas Research Network: The molecular taxonomy of primary prostate cancer, *Cell* 163:1011-1025, 2015. [Landmark paper providing the first genome-wide view of driver genetic events in prostate carcinoma].
- Epstein JI, Amin MB, Fine SW, et al: The 2019 Genitourinary Pathology Society (GUPS) White Paper on Contemporary Grading of Prostate Cancer, *Arch Pathol Lab Med* 145:461-493, 2021. [Current recommendations for prostate cancer grading].
- Litchfield K, Levy M, Huddart RA et al: The genomic landscape of testicular germ cell tumours: from susceptibility to treatment, *Nat Rev Urol* 13:409-419, 2016. [Discussion of the pathophysiology of testicular germ cell tumors and implications for therapy].
- Looijenga LH, Stoop H, Biermann K: Testicular cancer: biology and biomarkers, *Virchows Arch* 464:301-313, 2014. [Review of salient features of germ cell tumor biology and associated biomarkers].
- Menon S, Moch H, Berney DM, et al: WHO 2022 classification of penile and scrotal cancers: updates and evolution, *Histopathology* 82:508-520, 2023. [Latest classification system for penile neoplasms].
- Moch H, Amin MB, Berney DM, et al: The 2022 World Health Organization Classification of Tumours of the Urinary System and Male Genital Organs-Part A: Renal, Penile, and Testicular Tumours, *Eur Urol* 82:458-468, 2022. [Latest classification system for testicular neoplasms].
- Netto GJ, Amin MB, Berney DM et al: The 2022 World Health Organization Classification of Tumors of the Urinary System and Male Genital Organs-Part B: Prostate and Urinary Tract Tumors, *Eur Urol* 82:469-482, 2022. [Update on the classification of bladder carcinoma].
- Robertson AG, Kim J, Al-Ahmadie H et al: Comprehensive molecular characterization of muscle-invasive bladder cancer, *Cell* 171:540-566.e25, 2017. [Landmark paper providing integrated genomic and transcriptomic analysis of the driver genetic alterations of bladder carcinoma and implications for classification and treatment].
- Taylor-Weiner A, Zack T, O'Donnell E et al: Genomic evolution and chemoresistance in germ-cell tumours, *Nature* 540:114-118, 2016. [A landmark paper describing genetic drivers in testicular germ cell tumors and additional genomic changes that emerge with resistance to chemotherapy].